

BPS DENDROSCOPY ON LOCAL $\mathbb{F}_1$

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ABSTRACT. bla

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1. INTRODUCTION

Mapping out the spectrum of BPS states in type II string theories compactified on a Calabi-Yau threefold is an important problem in mathematical physics, with applications to supersymmetric black holes, supersymmetric gauge theories, algebraic geometry, representation theory and more. In previous works [1, 2], we studied this problem for two of the simplest non-trivial toric Calabi-Yau threefolds, the total spaces of the canonical bundle $X = K_S$ over the complex projective space $S = \mathbb{P}^2$ and the product of two lines $S = \mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$. Upon compactifying type IIA strings on K_S , these two geometries engineer rank 1 5D superconformal field theories on $\mathbb{R}_{3,1} \times S_1$, decoupled from gravity. The goal of this work is to carry out a similar study for the Hirzebruch surface $S = \mathbb{F}_1$, a non-trivial $\mathbb{P}^1$ bundle over $\mathbb{P}^1$ which arises by blowing up $\mathbb{P}^2$ at one point, and provides the UV completion of 5D $\mathcal{N} = 1$ super-Yang-Mills theory with gauge group $SU(2)$ and half-unit theta angle.

Unlike the (arguably trivial) case of the resolved conifold and other small resolutions of toric singularities, where the finite mass BPS spectrum consists only of mutually local D2-D0 brane bound states, the existence of a compact divisor S leads to a rich chamber structure for the Donaldson-Thomas (DT) invariants $\Omega_\sigma(\gamma)$ counting semi-stable objects in the derived category $\mathcal{C} = D^b \text{Coh}_c X$ of compactly supported sheaves for given stability condition $\sigma \in \text{Stab } X$ (Physically, $\Omega_\sigma(\gamma)$ reduces to the BPS index counting D4-D2-D0 bound states for given Kähler moduli on X along the slice of Π -stability conditions). The stability scattering diagram,

introduced first in the context of moduli spaces of quiver representations in [3], provides a powerful way to determine the DT invariants at any point $\sigma \in \text{Stab } X$, from the knowledge of the initial rays emitted at the boundary of $\text{Stab } X$ where certain spherical objects become massless. As anticipated in [4] and demonstrated in [5], this is the natural mathematical framework for the attractor flow tree formula [6, 7, 8, 9] allowing to express BPS indices at any point in moduli space, in terms of the indices $\Omega_\star(\gamma)$ counting BPS in their attractor chamber, also known as self-stability condition.

In previous work [1, 2], we constructed the scattering diagram for $S = \mathbb{P}^2$ and $S = \mathbb{F}_0$ both along the large volume slice (corresponding to divisorial stability conditions, generalizing the analysis of [10]), and along the physical slice of Π -stability conditions (determined by local mirror symmetry). In particular, the scattering diagram along the physical slice was manifestly invariant under the full group of autoequivalences (suitable extensions of the congruence subgroups $\Gamma_0(3)$ and $\Gamma_0(4)$, respectively, corresponding to the monodromy group of the family of elliptic curves). The case $S = \mathbb{F}_1$, associated by local mirror symmetry to the family of elliptic curves in $\mathbb{C}_x^\times \times \mathbb{C}_y^\times$ [11, 12, 13]

$$(1.1) \quad x + y + \frac{\ell}{xy} + \frac{\ell}{x} = \frac{1}{u},$$

where ℓ and u parametrized the gauge coupling and Coulomb branch parameter in the 5D gauge theory on $\mathbb{R}^{3,1} \times S_1$, is significantly more complicated for several reasons, which we believe warrant the present study:

- i) Due to the existence of effective curves with negative self-intersection (namely the base B of the $\mathbb{P}^1$ fibration, with $B^2 = -1$), line bundles are in general not everywhere stable in the large volume slice [14]. As result, the initial rays are not just pure D4 and anti-D4-branes (invertible sheaves $\mathcal{O}(d_F, d_B)$ and their homological shift $\mathcal{O}(d_F, d_B)[1]$, in mathematical terms) but also include spherical D2-branes wrapped on B . Nonetheless, upon extending the scattering diagram ‘below the parabola’ in affine coordinates, the initial rays turn out to be pure D4-anti-D4 brane pairs associated to a *geometric helix of invertible sheaves*. A similar phenomenon was observed for $S = \mathbb{F}_0$ in [2], and is expected to hold true for any del Pezzo surface with canonical polarization.
- ii) For generic values of the gauge coupling (or ‘mass parameter’) ℓ , the family (1.1) has the same ramification structure over the u -plane as for local $\mathbb{F}_0$, namely type $I_8 + 4I_1$ in Kodaira’s notation, where the I_8 fiber corresponds to $u = 0$ and the I_1 fibers to the four roots of the reduced discriminant

$$(1.2) \quad \Delta_4(u, \ell) = 1 + u - 8\ell u^2 - 36\ell u^3 + \ell(16\ell - 27)u^4.$$

In contrast to local $\mathbb{F}_0$, the Klein invariant $J(u, \ell)$ is in general not a Hauptmodul for any subgroup of $PSL(2, \mathbb{Z})$, which prevents us from unfolding the scattering diagram from the u plane to the Poincaré upper half-plane. Moreover, the fact that the discriminant is a general quartic polynomial in u (as opposed to a quadratic polynomial in u^2) makes it challenging to compute the periods at the four I_1 fibers, where we expect massless states to arise. Fortunately, we find that by passing to an 8:1 cover of the ℓ -plane, the roots can be expressed as rational functions in a new complex parameter¹ $b \in \mathbb{P}^1$,

$$(1.3) \quad \ell = -\frac{b(b^2 + b + 1)^3}{(b + 1)^8}, \quad u = -\frac{(b + 1)^4}{(b^2 + b + 1)(b^2 + 4b + 1)} \Rightarrow \Delta_4(u, \ell) = 0.$$

This change of variable naturally arises when computing the regulator period of nodal curves in the family (1.1) using the method of [15], allowing to express the periods at the conifold point (sometimes called quantum volume) as linear combinations of Bloch-Wigner dilogarithms in b and its powers, along with standard logarithms.

¹Simpler rational expressions exist in terms of $\beta = b + 1/b$, see (5.71), given a 4:1 cover of the ℓ -plane, but periods need not be invariant under $b \rightarrow 1/b$.

- iii) For the special value $\ell = -27/256$, two I_1 fibers collide to form a type II degeneration, known in physics as Argyres-Douglas (AD) point of type H_0 (in contrast, local $\mathbb{F}_0$ admits a degeneration $I_1 + I_1 \rightarrow I_2$, whose physics is simply unHiggsing). As a consequence, the monodromy group does reduce to (an extension) of a *non-congruence subgroup* $\Gamma \subset PSL(2, \mathbb{Z})$, a genus zero subgroup of index 10 with 3 cusps (corresponding to the I_8 and two I_1 fibers) and one elliptic point of order 3 (corresponding to the AD point) first identified in [16]. The Coulomb branch parameter u then becomes a Hauptmodul for Γ , allowing to parametrize the universal cover of the Coulomb branch by a modulus τ in the Poincaré upper half-plane. As in the case of local $\mathbb{P}^2$ and local $\mathbb{F}_0$ at the special point $\lambda = 1$, this allows to construct full Γ -invariant scattering diagram avoiding the need for branch cuts.
- iv) For the other special value $\ell = 0$, one I_1 fiber collides with the I_8 fiber at $u = 0$ to produce an I_9 fiber, resulting in the same ramification structure as for local $\mathbb{P}_2$. Indeed, geometrically ℓ controls the size of the exceptional divisor when viewing $\mathbb{F}_1$ as the one-point blow-up of $\mathbb{P}^2$. Since this degeneration changes the large volume structure, we refrain from studying it here.

The rest of this work is organized as follows...

Relevant math work: [17]

Acknowledgements: We are grateful to David Berghaus, Pierre Lairez, Bruno Le Floch, Marcos Marino, Marc Mezzarobba and Thorsten Schimannek for useful communications. This research was supported by Agence Nationale de la Recherche under contract number ANR-21-CE31-0021. *For the purpose of Open Access, a CC-BY public copyright licence has been applied by the authors to the present document and will be applied to all subsequent versions up to the Author Accepted Manuscript arising from this submission.*

2. GENERALITIES

2.1. Coherent sheaves on $K_{\mathbb{F}_1}$. The Hirzebruch surface $S = \mathbb{F}_1$ is a $\mathbb{P}^1$ bundle over $\mathbb{P}^1$, given by the projectivization of the $\mathcal{O} + \mathcal{O}(-1)$ bundle over $\mathbb{P}^1$. We denote by F and B the fiber and base $\mathbb{P}^1$, respectively. They satisfy $F^2 = 0, B^2 = -1, B.F = 1$. Alternatively, one may view $\mathbb{F}_1$ as the blow-up of $\mathbb{P}^2$ at one point. The second homology is then spanned by the hyperplane class $H = F + B$ and the exceptional divisor C , with $H^2 = 1, C^2 = -1, H.C = 0$. Although the exceptional divisor class C coincides with the base divisor class B , we use different letters to distinguish the bases (F, B) and (H, C) . The Mori cone is spanned by (F, B) , while the nef cone is spanned by (H, F) . The anticanonical class is $c_1(S) = 3F + 2B = 3H - C$. For any coherent sheaf E on S , we decompose the first Chern class as $c_1(E) = d_F(E)F + d_B(E)B$ and denote the Chern character by $\gamma(E) = [r(E), d_F(E), d_B(E), \text{ch}_2]$. Alternatively, we can decompose $c_1(E) = d_H(E)H + d_C(E)C$, with

$$(2.1) \quad (d_H, d_C) = (d_B, d_B - d_F) \Leftrightarrow (d_F, d_B) = (d_H, d_H + d_C)$$

We denote by $\mathcal{O}(m_F, m_B)$ the unique line bundle on S with first Chern class $m_F F + m_B B$; its Chern vector is $[1, m_F, m_B, m_F m_B - \frac{1}{2}m_B^2]$. Under tensoring $E \mapsto E \otimes \mathcal{O}(m_F, m_B)$, an operation we call spectral flow, the Chern vector transforms as

$$(2.2) \quad [r, d_F, d_B, \text{ch}_2] \mapsto [r, d_F + r m_F, d_B + r m_B, \text{ch}_2 + (d_F - d_B)m_B + d_B m_F + r(m_B m_F - \frac{1}{2}m_B^2)]$$

For two coherent sheaves E, E' of Chern characters $[r, d_B, d_F, \text{ch}_2]$ and $[r', d'_B, d'_F, \text{ch}'_2]$, the Euler form is given by the Grothendieck-Riemann-Roch theorem,

$$(2.3) \quad \begin{aligned} \chi(E, E') &:= \dim \text{Hom}(E, E') - \dim \text{Ext}^1(E, E') + \dim \text{Ext}^2(E, E') \\ &= r r' + \frac{1}{2} r (2d'_F + d'_B) - \frac{1}{2} r' (2d_F + d_B) + r \text{ch}'_2 + r' \text{ch}_2 - (d_F d'_B - d_B d'_F) + d_B d'_B \end{aligned}$$

We define the antisymmetrized Euler form, or Dirac product, as

$$(2.4) \quad \langle \gamma, \gamma' \rangle := \chi(E, E') - \chi(E', E) = r(2d'_F + d'_B) - r'(2d_F + d_B)$$

The Euler form and Dirac product are invariant under spectral flow. For $r \neq 0$, so is the Bogomolov discriminant, defined by

$$(2.5) \quad \Delta(\gamma) = \frac{2d_F d_B - d_B^2}{2r^2} - \frac{\text{ch}_2}{r}$$

2.2. Moduli space of Gieseker-semistable sheaves. Up to scaling, we can parametrize ample divisors D by a real number M , such that $D \propto H_M := 3F + (M+2)B = 3H + (M-1)C$. For $M = 0$, H_0 coincides with the anticanonical class $c_1(S)$. We define the slope by

$$(2.6) \quad \mu_M(\gamma) = \frac{3d_H + d_C}{r} - M \frac{d_C}{r},$$

for $r \neq 0$, and $\mu_M(\gamma) = +\infty$ for $r = 0$ and denote by $\mathcal{M}_M(\gamma)$ the moduli space of Gieseker-semistable sheaves with Chern character γ . When $\mathcal{M}_M$ is non empty, it has complex dimension

$$(2.7) \quad d_{\mathcal{C}}(\gamma) = 1 - \chi(E, E) = 2d_B d_F - d_B^2 - 2r \text{ch}_2 - r^2 + 1$$

We denote by $\Omega_M(\gamma)$ the Donaldson-Thomas invariant associated to the moduli space (or moduli stack, if γ is not primitive of M not generic) $\mathcal{M}_M(\gamma)$. For $r > 0$, the non-emptiness of $\mathcal{M}_M(\gamma)$ requires that the Bogomolov discriminant is positive, $\Delta(\gamma) \geq 0$. Sufficient conditions for $\mathcal{M}_M(\gamma)$ being non-empty are discussed in [18, 19, 20]. As in the classic case of the projective plane [?], these conditions are determined from the list of stable exceptional bundles E , with $\dim \text{Hom}(E, E) = 1, \dim \text{Ext}^{1,2}(E, E) = 0$. The list of the first few exceptional bundles, and their stability range, can be found in [18, Table 2],

$$(2.8) \quad [1, 0, 0, 0]_0, [2, 1, 1, -\frac{1}{2}]_{\frac{3}{8}}, [4, 2, 1, -\frac{3}{2}]_{\frac{15}{32}}, [5, 2, 2, -2]_{\frac{12}{25}}, [6, 3, 5, -\frac{5}{2}]_{\frac{35}{72}}, [8, 4, 1, -\frac{7}{2}]_{\frac{63}{128}}, \dots$$

where we wrote the discriminant $\Delta = \frac{1}{2}(1 - \frac{1}{r^2})$ in subscript. All exceptional bundles are stable for the canonical polarization $M = 0$, but (except for the first of them) get destabilized outside some finite range $[M_-, M_+]$. The list $\mathcal{E}_M$ of stable exceptional bundles determines a lower bound $\Delta(E) \geq \delta_{\text{DLP}}(M, a, b)$ on the discriminant of non-exceptional H -stable torsion-free sheaves with slope $(a, b) = (d_B, d_F)/r$, given by [18]²

$$(2.9) \quad \delta_{\text{DLP}}(M, a, b) := \sup_{F \in \mathcal{E}_M, |\mu_M(F) - (a+2b) - M(b-a)| \leq 2\frac{M+8}{M+2}} \left[P_{\epsilon} \left(a - \frac{d_B(F)}{r(F)}, b - \frac{d_F(F)}{r(F)} \right) - \Delta(F) \right]$$

where $P_{\epsilon}(a, b) = (\epsilon a + 1)(\epsilon b + 1 - \frac{1}{2}\epsilon a)$ and $\epsilon(F) = \text{sign}[\mu_M(F) - (a+2b) - M(b-a)] \leq 2\frac{M+8}{M+2}$. Note that $\delta_{\text{DLP}}(M, a, b)$ is invariant under $(a, b) \rightarrow (a+1, b), (a, b+1), (-a, -b)$ and greater than $1/2$. Conversely, $\mathcal{M}_H(\gamma)$ is non-empty if γ is the Chern character of an exceptional stable bundle or $\Delta(\gamma) > \delta_{\text{DLP}}(M, d_B/r, d_F/r)$. The function $\delta_{\text{DLP}}(M, a, b)$ for the canonical polarization $M = 0$ is plotted in Figure 1.

2.3. Gopakumar-Vafa invariants . For rank 0, Gieseker indices $[0, d_F, d_B, \text{ch}_2]$ are conjecturally independent of M and of ch_2 , and reduce to the genus 0 GV invariants $\text{GV}_{d_F, d_B}^{(0)}$. For $S = \mathbb{F}_1$, the latter were computed in [11, Table 10]:

$$(2.10) \quad \begin{array}{c|ccccc} d_F \backslash d_B & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & -2 & 3 & 0 & 0 & 0 \\ 2 & 0 & 5 & -6 & 0 & 0 \\ 3 & 0 & 7 & -32 & 27 & 0 \\ 4 & 0 & 9 & -110 & 286 & -192 \end{array}$$

They vanish unless $d_F \geq 0, d_B \geq 0$ and $d_B \leq d_F$, except for the entry $(d_F, d_B) = (0, 1)$, which lies outside the cone of moveable curves. The fact that they vanish for $d_B > 2d_F$ is a consequence

²The parameter m in [18] corresponds to $\frac{1-M}{2+M}$ in our notations.

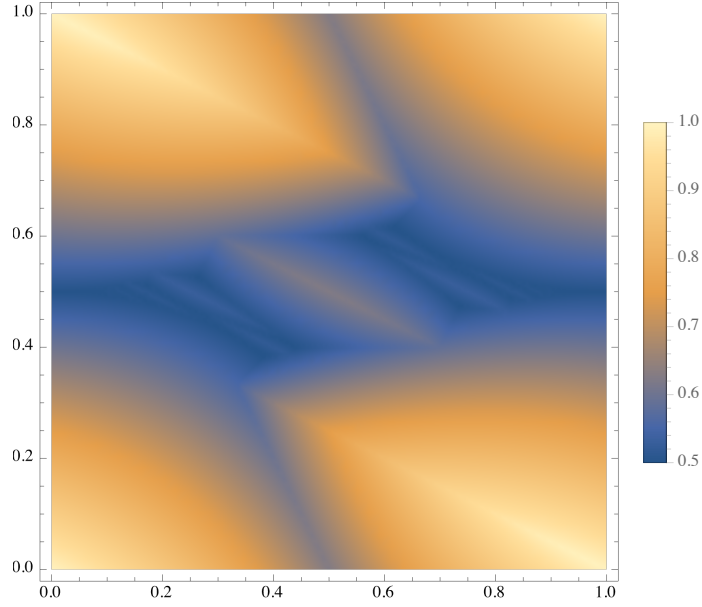


FIGURE 1. Lower bound $\delta_{\text{DLP}}(M, a, b)$ on the discriminant of non-exceptional H_M -stable sheaves for $M = 0$, obtained by retaining exceptional sheaves with rank up to 19. Figure adapted from [18, Fig. 6].

of the formula for the expected dimension of the moduli space, $\dim_{\mathbb{C}} \mathcal{M}_M(\gamma) = 2d_B d_F - d_B^2 + 1$. The vanishing in the range $d_F < d_B < 2d_F$ is instead a consequence of the existence of second geometric phase related by a flop $(T_B, T_F) \mapsto (-T_B, T_F + T_B)$. After flop, the GV invariants become $\text{GV}'_{d'_F, d'_B} = \text{GV}'_{d_F, d_F - d_B}$ for $d_F \geq d_B$ and zero otherwise, except for $\text{GV}'_{0,1} = \text{GV}_{0,1}^{(0)}$:

(2.11)

$d'_F \backslash d'_B$	0	1	2	3	4
0	0	1	0	0	0
1	3	-2	0	0	0
2	-6	5	0	0	0
3	27	-32	7	0	0
4	-192	286	-110	9	0
5	1695	-3038	1651	-288	11

In particular, the diagonal invariants $\text{GV}_{d,d}^{(0)}$, or the equivalently the invariants $\text{GV}'_{d,0}^{(0)}$ in the first column of the previous table, are recognized as the GV invariants of local $\mathbb{P}_2$.

The refined BPS invariants $\text{GV}_{d_B, d_F}^{(j_L, j_R)}$, counting multiplicities of representations of the 5D little group $SU(2)_L \times SU(2)_R$, were found in [21, p34], see also [13, Table 5]. The refined D2-D0 invariants follow from the refined BPS invariants by evaluating the character of the $SU(2)_L \times SU(2)_R$ representation on the diagonal locus $y_L = y_R$,

$$\begin{aligned}
 \Omega([0, 0, 1, \text{ch}_2]) &= 1 \\
 \Omega([0, 1, 0, \text{ch}_2]) &= -y - 1/y \\
 \Omega([0, 1, 1, \text{ch}_2]) &= y^2 + 1 + 1/y^2 \\
 \Omega([0, 2, 1, \text{ch}_2]) &= y^4 + y^2 + 1 + 1/y^2 + 1/y^4 \\
 \Omega([0, 2, 2, \text{ch}_2]) &= -y^5 - y^3 - y - 1/y - 1/y^3 - 1/y^5 \\
 \Omega([0, 3, 2, \text{ch}_2]) &= -y^9 - 3y^7 - 4y^5 - 4y^3 - 4y - \dots \\
 \Omega([0, 3, 3, \text{ch}_2]) &= y^{10} + 2y^8 + 3y^6 + 3y^4 + 3y^2 + 3 + \dots \\
 \Omega([0, 4, 2, \text{ch}_2]) &= -y^{13} - 3y^{11} - 8y^9 - 10y^7 - 11y^5 - 11y^3 - 11 + \dots
 \end{aligned}
 \tag{2.12}$$

The first and second line are straightforward consequences of the fact that the base curve B is isolated, while the fiber comes in a $\mathbb{P}^1$ family.

2.4. Bridgeland stability conditions. Let $\mathcal{C}$ be the derived category of compactly supported coherent sheaves on $K_{\mathbb{F}_1}$. For a bounded complex of sheaves $\mathcal{E} = \cdots \rightarrow E_k \rightarrow E_{k+1} \cdots$, we define the Chern vector as the alternating sum $\gamma(\mathcal{E}) = \sum_k (-1)^k \gamma(E_k)$. The image of the Grothendieck group $K(\mathcal{C})$ under the Chern map is a rank 4 lattice Λ , corresponding physically to the lattice of D4-, D2- and D0-brane charges. The Donaldson-Thomas invariant $\Omega_\sigma(\gamma)$, counting BPS states of fixed charges γ , depends on a choice of stability condition $\sigma = (Z, \mathcal{A})$, consisting of a linear map $Z : \Lambda \rightarrow \mathbb{C}$ (the central charge), and an abelian subcategory $\mathcal{A} \subset \mathcal{C}$ (the heart) satisfying certain axioms. As shown in [22], the space of stability conditions $\text{Stab } \mathcal{C}$ is a complex manifold of complex dimension $\text{rk } \Lambda = 4$. It admits commuting actions by auto-equivalences $\text{Aut } \mathcal{C}$ (from the left) and by the universal cover of $GL(2, \mathbb{R})^+$ (from the right). Up to a $\mathbb{C}^\times \subset GL(2, \mathbb{R})^+$ action, one can assume that D0-branes (or skyscraper sheaves) have central charge -1 , such that the central charge takes the form

$$(2.13) \quad Z(\gamma) = -rT_D + d_F T_F + d_B T_B - \text{ch}_2$$

The parameters T_B and T_F are complexified Kähler parameters measuring the size of the base and the fiber, such that the complexified Kähler form is $J_{\mathbb{C}} = T_B F + T_F (B + F)$. The Kahler cone condition is $\Im T_B, \Im T_F > 0$. The parameter T_D corresponds to the central charge of a D4-brane wrapped on S . Under tensoring by $\mathcal{O}(m_F, m_B)$, such that the Chern vector transform as in (2.2), the central charge remains invariant provided

$$(2.14) \quad (T_F, T_B, T_D) \mapsto (T_F + m_B, T_B + m_F - m_B, T_D + m_F T_F + m_B T_B + m_B m_F - \frac{1}{2} m_B^2)$$

Furthermore, it is invariant Under derived duality provided T_F, T_B, T_D are suitably conjugated,

$$(2.15) \quad Z_{-\overline{T_F}, -\overline{T_B}, \overline{T_D}}(-r, d_F, d_B, -\text{ch}_2) = -\overline{Z_{T_F, T_B, T_D}(r, d_F, d_B, \text{ch}_2)}$$

On the physical slice, T_D is determined by T_F and T_B by local mirror symmetry. In the large volume limit, the physical central charge reduces to $Z_{\text{LV}}(\gamma) = -\int_S e^{-J_{\mathbb{C}}} \text{ch}(E) \text{Tod}(S)$, up to exponentially small corrections, leading to a quadratic function of T_F and T_B , (consistent with [23, (4.34)] upon setting $m_1 = T_F, m_2 = T_B$)

$$(2.16) \quad T_D = \frac{1}{2} T_F^2 + T_B T_F + \frac{1}{6}$$

The canonical polarization, such that $\Im J_{\mathbb{C}} \propto c_1(S)$, corresponds to $\Im T_F = 2\Im T_B$. In fact, on the physical slice the combination $T_F - 2T_B$ turns out to be directly expressible in terms of the coefficients of the mirror curve (i.e. it has a trivial mirror map), so it is natural to define

$$(2.17) \quad T_F = 2T + m, \quad T_B = T - m$$

such that, in the large volume limit $T \rightarrow i\infty$ the central charge (2.16) becomes

$$(2.18) \quad Z(\gamma) = -r \left(T(4T + m) - \frac{m^2}{2} + \frac{1}{6} \right) + d_F(2T + m) + d_B(T - m) - \text{ch}_2$$

In this parametrization, the complexified Kähler form is $J_{\mathbb{C}} = T(3F + 2B) + mB$, and its imaginary part is proportional to H_M with $M = \Im m / \Im T$. Invariance under spectral flow is maintained provided

$$(2.19) \quad (T, m) \mapsto \left(T + \frac{1}{3} m_F, m + m_B - \frac{2}{3} m_F \right)$$

In particular, tensoring with the anticanonical class $\mathcal{O}(3F + 2B)$ leaves m invariant but shifts $T \mapsto T + 1$. Note that the flop acts by

$$(2.20) \quad (T, m) \mapsto \left(\frac{2T + m}{3}, \frac{5T - 2m}{3} \right)$$

leaving the locus $T = m$ invariant.

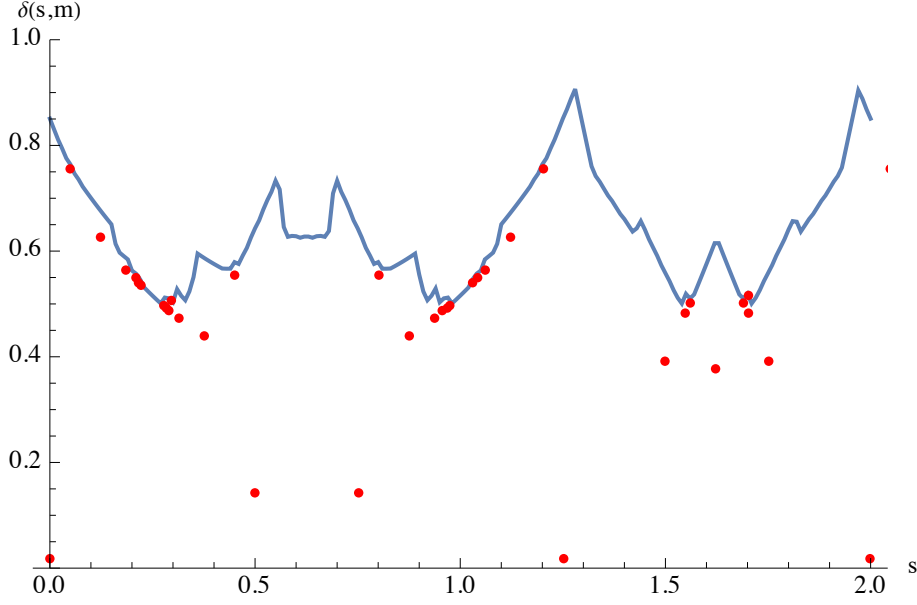


FIGURE 2. Approximate shape of $\delta(s, m)$ for $m = 1/4$. The red dots indicate the corresponding lowest values arising from exceptional stable bundles.

As in the case of $\mathbb{F}_0$, the construction of stability conditions depends on To fix

$$(2.21) \quad \delta(s, m) := \inf_E \left[\Delta(E) + \frac{1}{4} \left(\frac{d_H(E) + 3d_C(E)}{2r(E)} + m \right)^2 \right]$$

where the infimum runs over slope-stable bundles E with slope $s = \frac{3d_H(E) + d_C(E)}{4r(E)}$, or equivalently $s = \frac{d_B + 2d_F}{4r}$. The function $\delta(\mu, m)$ is related to $\delta_{\text{DLP}}(0, a, b)$,

$$(2.22) \quad \delta(s, m) = \inf_{\frac{a}{4} + \frac{b}{2} = s} \left[\delta_{\text{DLP}}(0, a, b) + \frac{1}{4} \left(\frac{3a - 2b}{2} + m \right)^2 \right]$$

see Figure 2.

3. QUIVER DESCRIPTIONS

In this section, we introduce three exceptional collections on $S = \mathbb{F}_1$, which give rise to three equivalent quiver descriptions of the category $\mathcal{C} = D^b \text{Coh}(S)$. As shown in [14, 24], these collections play a distinguished role in studying the stability of line bundles and proving that the moduli space of semi-stable objects are projective. In the next Section, we will show that these quiver descriptions are sufficient to determine all incoming rays in the large volume scattering diagram.

We start by recalling some definitions and basic facts. For a given projective surface S , an exceptional collection is an ordered list $\mathcal{E} = (E^1, \dots, E^n)$ of coherent sheaves such that

$$(3.1) \quad \text{Hom}(E^i, E^i) \simeq \mathbb{C}, \quad \text{Ext}^k(E^i, E^j) = 0 \quad \forall k \geq 0, 1 \leq j \leq i \leq r$$

In particular, the objects E_i are exceptional sheaves. $\mathcal{E}$ is a *strongly* exceptional collection if for all i, j , $\text{Ext}^k(E^i, E^j) = 0$ whenever $k > 0$. The matrix $S^{ij} = \chi(E^i, E^j)$ is then upper triangular with ones along the diagonal and non-negative entries. It is a *full* exceptional collection if the objects span the K-theory lattice; for a simply connected surface, this implies $r = \chi(S) = b_2(S) + 2$. In order to generate the category $\mathcal{C}$, we need the following stronger condition, sometimes called *cyclicity*: let $\mathcal{H} = \{E^i\}_{i \in \mathbb{Z}}$ be the bi-infinite sequence of coherent sheaves defined by the condition $E^{i+r} = E^i \otimes K_S^{-1}$. We require that $\mathcal{H}$ forms a *geometric helix*, i.e. that each finite sequence $(E_i, E_{i+1}, \dots, E_{i+r})$ of length r (called foundation of the helix) is a full exceptional collection and that

$$(3.2) \quad \forall i < j, \quad \text{Ext}^k(E^i, E^j) = 0 \quad \text{unless} \quad k = 0$$

We then define the dual collection $\mathcal{E}^\vee = (E_1, \dots, E_r)$ of objects in $D^b \text{Coh}(S)$ by

$$(3.3) \quad E_i = L_{E^1} L_{E^2} \dots L_{E^{i-1}} E^i$$

where $L_E F$ denotes the left mutation of F with respect to E . In particular, it follows that $\chi(E^i, E_j) = \delta_j^i$, and $S_{ij} := \chi(E_j, E_i)$ (equal to the inverse of S^{ij}) is an upper triangular matrix with ones along the diagonal (but other entries may be negative). The dual collection $\mathcal{E}^\vee$ is exceptional but in general not strongly so. Nonetheless, it generates the derived category $\mathcal{C}$, in the sense that the latter is isomorphic to the category of representation $D^b(Q, R)$ of a quiver with potential, where the vertices $i \in Q_0$ correspond to the dual objects E_i , the arrows $(i \rightarrow j) \in Q_1$ to the negative entries $S_{ij} < 0$, and the relations R_a to the positive off-diagonal entries. Equivalently, it is isomorphic to the category of representation $D^b(Q', W)$ of a quiver with potential, where the quiver Q' includes additional arrows $a : (j \rightarrow i)$ for any off-diagonal positive entry $S_{ij} > 0$, and a (super)potential $W = \sum_a \phi_a R_a$ linear in the additional arrows.

3.1. Bridgeland-Stern-Perling collection. The simplest exceptional collections come from geometric helices of invertible sheaves (i.e. rank 1 sheaves with vanishing discriminant). In the case of $S = \mathbb{F}_1$, there is a unique geometric helix, up to translation of the index i and tensoring by $\mathcal{O}(m_F F + m_B B)$ [25, 26, Example 8.6.(a)]

$$(3.4) \quad \dots, \mathcal{O}(-F - B), \mathcal{O}, \mathcal{O}(F), \mathcal{O}(F + B), \mathcal{O}(2F + B), \mathcal{O}(3F + 2B), \mathcal{O}(4F + 2B), \dots$$

first constructed in [27, p. 100], as quoted in [28, §4.1]. Choosing the foundation

$$(3.5) \quad \mathcal{C} = (\mathcal{O}, \mathcal{O}(F), \mathcal{O}(F + B), \mathcal{O}(2F + B)) = (\mathcal{O}, \mathcal{O}(H - C), \mathcal{O}(H), \mathcal{O}(2H - C))$$

we arrive at the dual Ext-exceptional collection $\mathcal{C}^\vee = (E_1, E_2, E_3, E_4)$ with

$$(3.6) \quad E_1 = \mathcal{O}, \quad E_2 = \mathcal{O}(-F)[1], \quad E_3 = \mathcal{O}(-B)[1], \quad E_4 = \mathcal{O}(-B - F)[2]$$

The Chern characters $\gamma^i = \text{ch } E^i$, $\gamma_i = \text{ch } E_i$ of the corresponding objects are given, in components $[r, d_F, d_B, \text{ch}_2]$ by [CollDBridgeland](#) and [CollBridgeland](#), respectively

$$(3.7) \quad \begin{array}{ll} \gamma^1 = [1, 0, 0, 0] & \gamma_1 = [1, 0, 0, 0] \\ \gamma^2 = [1, 1, 0, 0] & \gamma_2 = [-1, 1, 0, 0] \\ \gamma^3 = [1, 1, 1, \frac{1}{2}] & \gamma_3 = [-1, 0, 1, \frac{1}{2}] \\ \gamma^4 = [1, 2, 1, \frac{3}{2}] & \gamma_4 = [1, -1, -1, \frac{1}{2}] \end{array}$$

The resulting quiver is shown in Figure 3 (left), accompanied by the superpotential [29, 30]³

$$(3.8) \quad W = -\Phi_{23}\Phi_{34}^1\Phi_{41}^2\Phi_{12}^2 + \Phi_{23}\Phi_{34}^2\Phi_{41}^2\Phi_{12}^1 + \Phi_{24}\Phi_{41}^1\Phi_{12}^2 - \Phi_{24}\Phi_{41}^3\Phi_{12}^1 + \Phi_{34}^1\Phi_{41}^3\Phi_{13} - \Phi_{34}^2\Phi_{41}^1\Phi_{13}$$

A general Chern character $\gamma = [r, d_F, d_B, \text{ch}_2]$ decomposes as $\gamma = \sum_i n_i \gamma_i$ with

$$(3.9) \quad \begin{aligned} n_1 &= d_F + \frac{1}{2}d_B + \text{ch}_2 + r, & n_2 &= d_F - \frac{1}{2}d_B + \text{ch}_2 \\ n_3 &= \frac{1}{2}d_B + \text{ch}_2, & n_4 &= -\frac{1}{2}d_B + \text{ch}_2 \end{aligned}$$

In order that the dimension vector be positive for large negative ch_2 , we set $N_i = -n_i$. In the chamber $\zeta_1 \leq 0, \zeta_2 \geq 0$, such that the arrows Φ_{12}, Φ_{13} are set to zero, the dimension of the moduli space of stable quiver representations of dimension vector $\vec{N}$ agrees with the expected dimension of the moduli space of stable sheaves with Chern character γ .

Since the kernel of the antisymmetrized Euler form is spanned by the dimension vectors $(1, -1, 2, 0)$ and $(0, 2, -1, 1)$ (which sum up to the D0-brane charge $(1, 1, 1, 1)$), it is natural to restrict the scattering diagram to the two-dimensional slice

$$(3.10) \quad \zeta_1 - \zeta_2 + 2\zeta_3 = -\frac{1}{2} = 2\zeta_2 - \zeta_3 + \zeta_4$$

³Note that the same quiver arises for the strong exceptional collection $\mathcal{C}_0 = (\mathcal{O}, \mathcal{O}(B), \mathcal{O}(B + F), \mathcal{O}(2B + 2F)) = (\mathcal{O}, \mathcal{O}(C), \mathcal{O}(H), \mathcal{O}(2H))$, mentioned in [31, (5.6)] and used in [23, §4.2]. $\mathcal{C}_0$ arises as a foundation of the helix (3.4) tensored by $\mathcal{O}(-F)$. While the quivers are identical, the dimension vectors for fixed Chern character will be different.

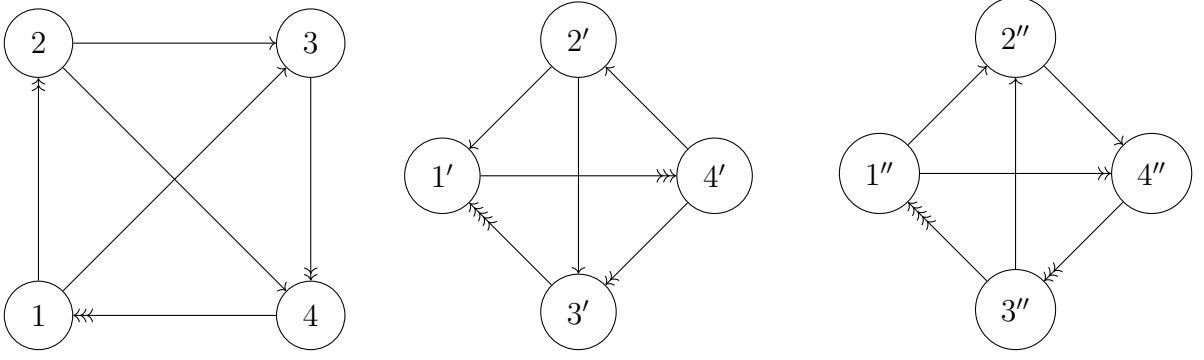


FIGURE 3. Quivers corresponding to the exceptional collections $\mathcal{C}, \mathcal{C}', \mathcal{C}''$. Observe that the usual quiver for $S = \mathbb{P}^2$ arises by merging the nodes 2 and 3 in the first quiver (a procedure called Higgsing). This is possible because there is only one arrow from 2 to 3,

We parametrize it as

$$(3.11) \quad \zeta_1 = v - 2u, \quad \zeta_2 = v - \frac{1}{2}, \quad \zeta_3 = u - \frac{1}{2}, \quad \zeta_4 = u - 2v,$$

The geometric rays are then lines

$$(3.12) \quad (N_3 + N_4 - 2N_1)u + (N_1 + N_2 - 2N_4)v - \frac{1}{2}(N_2 + N_3) = 0$$

oriented along $v_\gamma = (2N_4 - N_1 - N_2, N_3 + N_4 - 2N_1)$. We conjecture that the only initial rays are those associated to γ_i , with BPS index $\Omega_\star(\gamma_i) = 1$, and to the D0-branes $k\delta$ with $\delta = (1, 1, 1, 1)$, with unspecified BPS index $\Omega_\star(k\delta)$.

We find the following trees:

$[r, d_F, d_B, \text{ch}_2]$	$\vec{N}$	Trees	$\Omega_c(\vec{N})$
$[1, 0, 0, -1]$	$(0, 1, 1, 1)$	$\{\gamma_2, \{\gamma_3, \gamma_4\}\}$ $\{\{\gamma_2, \gamma_4\}, \gamma_3\}$	$y^2 + 2 + 1/y^2$
$[1, 0, 0, -2]$	$(1, 2, 2, 2)$	$\{\{\{\gamma_1, \{\{\{\gamma_2, \gamma_4\}, \gamma_3\}, \gamma_2\}\}, \gamma_4\}, \gamma_3\}$ $\{\{\{\{\gamma_1, \gamma_2\}, 2\gamma_4\}, 2\gamma_3\}, \gamma_2\}$ $\{(\{\{\{\{\gamma_1, \gamma_4\}, \gamma_3\}, \{\{\gamma_2, \gamma_4\}, \gamma_3\}\}), \gamma_2\}$	$y^4 + 3y^2 + 6 + 3/y^2 + 1/y^4$
$[2, 0, 0, -2]$	$(0, 2, 2, 2)$	$\{2\gamma_2, \{2\gamma_3, 2\gamma_4\}\}$ $\{\{\{\{\gamma_2, \gamma_4\}, \gamma_3\}, \gamma_2\}, \{\gamma_3, \gamma_4\}\}$	$-y^5 - 2y^3 - 3y - \dots$
$[2, -1, -1, -\frac{3}{2}]$	$(1, 2, 2, 1)$	$\{\{2\gamma_2, \{2\gamma_3, \gamma_4\}\}, \gamma_1\}$ $\{\{\{2\gamma_2, \{\gamma_3, \gamma_4\}\}, \gamma_1\}, \gamma_3\}$	$y^4 + 3y^2 + 5 + \dots$
$[3, -1, -1, -\frac{3}{2}]$	$(0, 2, 2, 1)$	$\{2\gamma_2, \{2\gamma_3, \gamma_4\}\}$	$y^2 + 1 + 1/y^2$
$[3, 0, 0, -3]$	$(0, 3, 3, 3)$	$\{3\gamma_2, \{3\gamma_3, 3\gamma_4\}\}$ $\{2\gamma_2, \{\{\{\gamma_2, \gamma_4\}, \gamma_3\}, \{2\gamma_3, 2\gamma_4\}\}\}$	$y^{10} + 2y^8 + 5y^6 + 8y^4 + 10y^2 + 11$
$[3, -2, -2, -1]$	$(1, 2, 2, 0)$	$\{\{2\gamma_2, 2\gamma_3\}, \gamma_1\}$	$y^2 + 1 + 1/y^2$
$[4, -1, -1, -\frac{5}{2}]$	$(0, 3, 3, 2)$	$\{3\gamma_2, \{3\gamma_3, 2\gamma_4\}\}$ $\{\{2\gamma_2, \{\gamma_3, \gamma_4\}\}, \{\gamma_2, \{2\gamma_3, \gamma_4\}\}\}$	$y^6 + 2y^4 + 5y^2 + 6 + \dots$

3.2. Mutated collections. In [24, §5.1], the authors introduce two strong exceptional collections, which no longer consist of invertible sheaves. The first one $\mathcal{C}'$ is obtained by mutating (3.5) with respect to node 4 (with + sign). This leads to the objects **We need to move 4 in front of 2,3**

$$(3.13) \quad E'^1 = \mathcal{O}, \quad E'^2 = \mathcal{O}(F), \quad E'^3 = \mathcal{O}(F + B), \quad E'^4 = V_2$$

where V_2 is the rank 2 exceptional vector bundle of Chern class $B + F$, with dual objects

$$(3.14) \quad E'_1 = \mathcal{O}, \quad E'_2 = \mathcal{O}(B)|_B[1], \quad E'_3 = \mathcal{O}(-2F - B)[2], \quad E'_4 = \mathcal{O}(-F - B)[1]$$

Their respective Chern vectors are given by [CollDArcaraII](#) and [CollAracaraII](#), respectively

$$(3.15) \quad \begin{aligned} \gamma^1 &= [1, 0, 0, 0] & \gamma'_1 &= [1, 0, 0, 0] \\ \gamma^2 &= [1, 1, 0, 0] & \gamma'_2 &= \left[0, 0, -1, \frac{1}{2}\right] \\ \gamma^3 &= \left[1, 1, 1, \frac{1}{2}\right] & \gamma'_3 &= \left[1, -2, -1, \frac{3}{2}\right] \\ \gamma^4 &= \left[2, 1, 1, -\frac{1}{2}\right] & \gamma'_4 &= \left[-1, 1, 1, -\frac{1}{2}\right] \end{aligned}$$

and the quiver is shown in Fig. 3 (center).

The second strong exceptional collection $\mathcal{C}''$ is obtained from (3.15) by mutating with respect to node 2 (with + sign), leading to the objects [We need to exchange 3 and 4](#)

$$(3.16) \quad E''^1 = \mathcal{O}, \quad E''^2 = \mathcal{O}(B), \quad E''^3 = \mathcal{O}(F + B), \quad E''^4 = V_2$$

with dual objects

$$(3.17) \quad E''_1 = \mathcal{O}, \quad E''_2 = \mathcal{O}(B)|_B, \quad E''_3 = \mathcal{O}(-2F - B)[2], \quad E''_4 = \mathcal{O}(-F)[1]$$

Equivalently, $\mathcal{C}''$ can be obtained from the collection $\mathcal{C}_0$ mentioned in footnote 3 by mutating with respect to node 4 (with + sign). The Chern vectors of the respective objects are [CollDArcaraI](#) and [CollAracaraI](#), respectively

$$(3.18) \quad \begin{aligned} \gamma''^1 &= [1, 0, 0, 0] & \gamma''_1 &= [1, 0, 0, 0] \\ \gamma''^2 &= \left[1, 0, 1, -\frac{1}{2}\right] & \gamma''_2 &= \left[0, 0, 1, -\frac{1}{2}\right] \\ \gamma''^3 &= \left[1, 1, 1, \frac{1}{2}\right] & \gamma''_3 &= \left[1, -2, -1, \frac{3}{2}\right] \\ \gamma''^4 &= \left[2, 1, 1, -\frac{1}{2}\right] & \gamma''_4 &= [-1, 1, 0, 0] \end{aligned}$$

and the quiver is shown in Fig. 3 (right). Note that the D0-brane dimension vector is $(1, 1, 1, 2)$ for both $\mathcal{C}$ and $\mathcal{C}'$.

4. LARGE VOLUME SCATTERING DIAGRAM

In this section, we study the scattering diagram along the large volume slice, obtained by dropping the constant $1/6$ for the central charge (2.18):

$$(4.1) \quad Z^{\text{LV}}(\gamma) = -r \left(T(4T + m) - \frac{m^2}{2} \right) + d_F(2T + m) + d_B(T - m) - \text{ch}_2$$

For simplicity, we assume that m is a fixed real parameter, and aim to construct the scattering diagram in the (s, t) half-plane, where $T = s + it$. By spectral flow invariance, we may restrict m to a width 1 interval, for example $-1/3 < m < 2/3$. Moreover, the diagram is periodic under $s \mapsto s + 1$, with a suitable relabelling of the charges. As in previous studies, there is no loss in assuming that the phase $\psi = 0$, since other phases (distinct from $\pm\pi/2$) can be reached by transforming

$$(4.2) \quad (s, t) \mapsto \left(s - t \tan \psi, \frac{t}{\cos \psi} \right)$$

Note that this only holds for real m , and on the large volume slice. For $\psi = 0$, geometric rays for $r \neq 0$ are branches of hyperbolae centered at $s = \frac{d_F + 2d_B - mr}{8r}$ on the real axis, or for $r = 0$, vertical rays at $s = \frac{\text{ch}_2 + m(d_F - d_B)}{d_F + 2d_B}$. In particular, for pure D4-branes with flux, the geometric ray $\mathcal{R}(\mathcal{O}(d_F, d_B))$ starts at $s = \min(\frac{1}{4}(2d_F - d_B + m), \frac{1}{2}(d_B - m))$ on the real axis and bends to the left, while $\mathcal{R}(\mathcal{O}(d_F, d_B)[1])$ starts at $s = \max(\frac{1}{4}(2d_F - d_B + m), \frac{1}{2}(d_B - m))$ and bends to the right. Importantly, in contrast to the local $\mathbb{P}^2$ and local $\mathbb{F}_0$ cases studied in [1, 2], the structure sheaf and its shifts are *not* in general stable for any (s, t) , so the actual ray may not intersect the real axis.

We shall also be interested in the diagram in affine (x, y) coordinates defined (in analogy to [10]) by

$$(4.3) \quad x = \frac{\Re(e^{-i\psi}T)}{\cos \psi} = s, \quad y = -\frac{\Re(e^{-i\psi}T_D)}{\cos \psi}, \quad \mu = \frac{\Re(e^{-i\psi}m)}{\cos \psi} = m$$

where the upper half-plane $t > 0$ maps to the region above the parabola

$$(4.4) \quad P = \{(x, y), y = -4x^2 - \mu x + \frac{1}{2}\mu^2\}$$

In these coordinates, all rays are straight segments

$$(4.5) \quad ry + x(d_B + 2d_F) - \text{ch}_2 + \mu(d_F - d_B) = 0$$

where $\mu = m$ when m is real, the case of interest here. In order to regulate the scattering diagram (both in (s, t) and (x, y) plane), we use the cost function

$$(4.6) \quad \varphi_s(\gamma) = \frac{\Im Z^{\text{LV}}(\gamma)}{\Im T} = d_B + 2d_F - r(8s + m)$$

which coincides with $\Im Z^{\text{LV}}/\Im T$, and increases along the rays (with the orientation pointing away from the real axis).

In order to determine the structure of the initial rays along the boundary at $t = 0$, we can use the Ext-exceptional collections $\mathcal{C}, \mathcal{C}', \mathcal{C}''$ introduced in Eqs. (3.7), (3.15), (3.18) of the previous Section to rule out incoming rays from all but a finite set of points (in a finite width integral). In particular, assuming that $3m$ is not an integer, we find that for any fixed integer $m_F \in \mathbb{Z}$, there is a unique integer m_B such that the domain of validity of the collection $\mathcal{C} \otimes \mathcal{O}(m_F, m_B)$ includes an open integral on the axis $t = 0$: **Recheck labels**

$$(4.7) \quad \mathcal{C} : \quad m_B = \lceil \frac{2m_F - 1}{3} + m \rceil$$

$$(4.8) \quad \mathcal{C}' : \quad m_B = \lceil \frac{2m_F - 3}{3} + m \rceil,$$

$$(4.9) \quad \mathcal{C}'' : \quad m_B = \lceil \frac{2m_F - 1}{3} + m \rceil.$$

In this way, we find that the full real axis is covered by the quiver descriptions listed in Table 1 (see Figures 4, 5, 6 for their domain of validity extending into the upper half-plane $t > 0$). We conclude that the only rays entering through the real axis are those listed on the last column of Table 1. In particular, in sharp contrast to the local $\mathbb{P}^2$ and local $\mathbb{F}_0$ case, the pure D4-branes and anti-D4-branes (in other words, the structure sheaf $\mathcal{O}(d_F, d_B)$ and its homological shift $\mathcal{O}(d_F, d_B)[1]$) are in general *not* stable throughout the upper half plane $t > 0$, while the D2-D0 branes with charge $[0, 1, 0, n]$ and $\pm[0, 0, 1, n + \frac{1}{2}]$ for some $n \in \mathbb{Z}$ are stable throughout the upper half plane. Indeed, as shown in [14], most of pure D4 and anti D4-branes decay precisely by emitting a D2-D0 brane with charge $[0, 0, 1, n + \frac{1}{2}]$. This is consistent with their index being 1 at large t , since the latter D2-D0 brane, associated with the curve B with negative self-intersection, also has unit DT invariant $\text{GV}_{(0,1)}^{(0)} = 1$. This can be seen from the trees contributing the corresponding indices, displayed in Table 2.

In order to understand the origin of the infinite family of rays entering at $t = 0$, we can use the affine coordinates (x, y) and extend the scattering diagram below the parabola P . In this way we find

- i) the infinite bunches of rays emerging at $s \in -\frac{m}{2} + \mathbb{Z}$ originate from the scattering of a pair $\mathcal{O}(d_F, d_B), \mathcal{O}(d_F - 1, d_B)[1]$ of pure (anti) D4-branes with Dirac product ± 2 .
- ii) The corresponding bound states $k\mathcal{O}(d_F, d_B) + (k+1)\mathcal{O}(d_F - 1, d_B)[1]$ and $(k+1)\mathcal{O}(d_F, d_B) + k\mathcal{O}(d_F - 1, d_B)[1]$ account for the rays $\mathcal{O}(d_F + k, d_B)$ and $\mathcal{O}(d_F - k - 1, d_B)[1]$, while the D4-anti D4 bound state $\mathcal{O}(d_F, d_B) + \mathcal{O}(d_F - 1, d_B)[1]$ accounts for the D2-D0 bound state $\text{GV}(1, 0, d_B)$ with index $\text{GV}_{(1,0)}^{(0)} = -y - 1/y$
- iii) In contrast, the spherical D2-D0 branes $\text{GV}(0, 1, n + \frac{1}{2})$ entering at $s = n \in \mathbb{Z}$ arise from the scattering of two rays with Dirac product ± 1 intersecting below the parabola P .

As a result, all rays emerging at $t = 0$ arise from the scattering of D4-branes and anti D4-branes associated to the helix (3.4), or its tensor product with $\mathcal{O}(F + B)$ ($0 < m < \frac{1}{3}$) or with $\mathcal{O}(-F)$

$(\frac{1}{3} < m < \frac{2}{3})$:

$$(4.10) \quad \left| \begin{array}{l} -\frac{1}{3} < m < 0 \\ 0 < m < \frac{1}{3} \\ \frac{1}{3} < m < \frac{2}{3} \end{array} \right| \begin{array}{l} \dots, \mathcal{O}(-1, -1), \mathcal{O}(0, 0), \mathcal{O}(1, 0), \mathcal{O}(1, 1), \mathcal{O}(2, 1), \mathcal{O}(3, 2), \dots \\ \dots, \mathcal{O}(-1, 0), \mathcal{O}(0, 0), \mathcal{O}(1, 1), \mathcal{O}(2, 1), \mathcal{O}(2, 2), \mathcal{O}(3, 2), \dots \\ \dots, \mathcal{O}(-2, -1), \mathcal{O}(0, 0), \mathcal{O}(0, 1), \mathcal{O}(1, 1), \mathcal{O}(2, 2), \mathcal{O}(3, 2), \dots \end{array} \right|$$

Dirac products of consecutive elements of the helix are $\kappa = 1, 2, 3, 2, \dots$. Collisions with $\kappa = 1, 2, 3$ occur below, at and above the parabola P , respectively, reproducing the above list of rays entering at $t = 0$ (see Figure 7). The initial point at which a pair $\pm\mathcal{O}(d_B, d_F)$ in the helix is emitted is irrelevant, provided it lies along the segment joining two consecutive intersections. For $|m|$ small enough, it can be chosen to be the point where the cost function (4.6) vanishes. In Section 5 we shall determine the exact location for the physical slice, corresponding to the values of (x, y) where the corresponding BPS state becomes massless.

m	$s_- < s < s_+$	Quiver	Incoming rays at s_+
$-\frac{1}{3} < m < 0$	$\frac{m-1}{4} < s < \frac{m}{4}$	$\mathcal{C}$	$\mathcal{O}(0, 0)$
	$\frac{m}{4} < s < -\frac{m}{2}$	$\mathcal{C}'' \otimes \mathcal{O}(1, 0)$	$\mathcal{O}(k+1, 0), \mathcal{O}(-k, 0)[1], \text{GV}(1, 0, 0)$
	$-\frac{m}{2} < s < \frac{m+1}{4}$	$\mathcal{C} \otimes \mathcal{O}(1, 1)$	$\mathcal{O}(1, 1)$
	$\frac{m+1}{4} < s < m + \frac{1}{2}$	$\mathcal{C}'' \otimes \mathcal{O}(2, 1)$	$\text{GV}(0, 1, \frac{1}{2})$
	$m + \frac{1}{2} < s < \frac{m+2}{4}$	$\mathcal{C}' \otimes \mathcal{O}(2, 1)$	$\mathcal{O}(1, 0)[1]$
	$\frac{m+2}{4} < s < \frac{1-m}{2}$	$\mathcal{C} \otimes \mathcal{O}(2, 1)$	$\mathcal{O}(k+2, 1), \mathcal{O}(1-k, 1)[1], \text{GV}(1, 0, 1)$
	$\frac{1-m}{2} < s < \frac{m+3}{4}$	$\mathcal{C}' \otimes \mathcal{O}(3, 2)$	$\mathcal{O}(2, 1)[1]$
$0 < m < \frac{1}{3}$	$-\frac{m}{2} < s < \frac{m}{4}$	$\mathcal{C}' \otimes \mathcal{O}(1, 1)$	$\mathcal{O}(0, 0)[1]$
	$\frac{m}{4} < s < \frac{m+1}{4}$	$\mathcal{C} \otimes \mathcal{O}(1, 1)$	$\mathcal{O}(1, 1)$
	$\frac{m+1}{4} < s < \frac{1-m}{2}$	$\mathcal{C}'' \otimes \mathcal{O}(2, 1)$	$\mathcal{O}(k+2, 1), \mathcal{O}(1-k, 1)[1], \text{GV}(1, 0, 1)$
	$\frac{1-m}{2} < s < \frac{m+2}{4}$	$\mathcal{C} \otimes \mathcal{O}(2, 2)$	$\mathcal{O}(2, 2)$
	$\frac{m+2}{4} < s < m + \frac{1}{2}$	$\mathcal{C}'' \otimes \mathcal{O}(3, 2)$	$\text{GV}(0, 1, \frac{1}{2})$
	$m + \frac{1}{2} < s < \frac{m+3}{4}$	$\mathcal{C}' \otimes \mathcal{O}(3, 2)$	$\mathcal{O}(2, 1)[1]$
	$\frac{m+3}{4} < s < 1 - \frac{m}{2}$	$\mathcal{C} \otimes \mathcal{O}(3, 2)$	$\mathcal{O}(k+3, 2), \mathcal{O}(2-k, 2)[1], \text{GV}(1, 0, 2)$
$\frac{1}{3} < m < \frac{2}{3}$	$-\frac{m}{2} < s < \frac{m-1}{4}$	$\mathcal{C} \otimes \mathcal{O}(0, 1)$	$\mathcal{O}(0, 1)$
	$\frac{m-1}{4} < s < m - \frac{1}{2}$	$\mathcal{C}'' \otimes \mathcal{O}(1, 1)$	$\text{GV}(0, 1, -\frac{1}{2})$
	$m - \frac{1}{2} < s < \frac{m}{4}$	$\mathcal{C}' \otimes \mathcal{O}(1, 1)$	$\mathcal{O}(0, 0)[1]$
	$\frac{m}{4} < s < \frac{1-m}{2}$	$\mathcal{C} \otimes \mathcal{O}(1, 1)$	$\mathcal{O}(1+k, 1), \mathcal{O}(-k, 1)[1], \text{GV}(1, 0, 1)$
	$\frac{1-m}{2} < s < \frac{m+1}{4}$	$\mathcal{C}' \otimes \mathcal{O}(2, 2)$	$\mathcal{O}(1, 1)[1]$
	$\frac{m+1}{4} < s < \frac{m+2}{4}$	$\mathcal{C} \otimes \mathcal{O}(2, 2)$	$\mathcal{O}(2, 2)$
	$\frac{m+2}{4} < s < 1 - \frac{m}{2}$	$\mathcal{C}'' \otimes \mathcal{O}(3, 2)$	$\mathcal{O}(k+3, 2), \mathcal{O}(2-k, 2)[1], \text{GV}(1, 0, 2)$

TABLE 1. Quiver descriptions at $t = 0$, obtained by tensoring one of the Ext-collections $\mathcal{C}$ (3.7), $\mathcal{C}'$ (3.15) and $\mathcal{C}''$ (3.18) with a suitable line bundle $\mathcal{O}(m_F, m_B)$. When both a D4-brane $\mathcal{O}(d_F, d_B)$ and an anti D4-brane $\mathcal{O}(d'_F, d'_B)[1]$ with $|\langle \gamma, \gamma' \rangle| = 2$ enter the boundary at the same point, they produce an infinite set of rays, indexed by $k \in \mathbb{N}$. Quiver descriptions outside the range $s_{\min} < s < s_{\min} + 1$ covered in the table can be obtained by tensoring with $\mathcal{O}(3k, 2k)$.

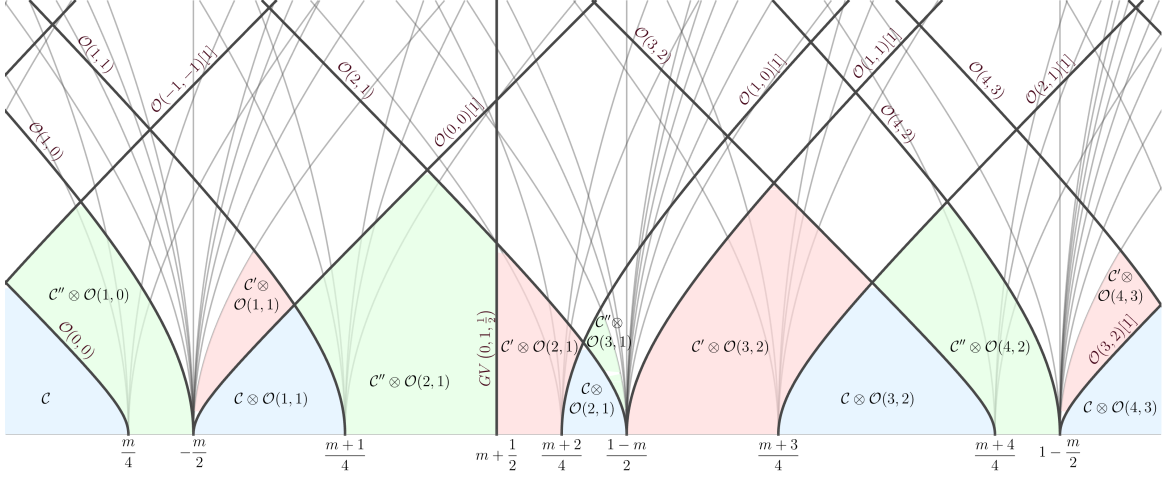


FIGURE 4. Domains of validity of the quiver descriptions covering the real axis, assuming that $-\frac{1}{3} < m < 0$ (plotted here for $m = -\frac{1}{10}$). Quivers obtain by tensoring the Ext-exceptional collections $\mathcal{C}$ (3.7), $\mathcal{C}'$ (3.15) and $\mathcal{C}''$ (3.18) with a suitable line bundle $\mathcal{O}(m_F, m_B)$ are plotted in blue, red and green, respectively. Grey lines correspond to the geometric rays for $\mathcal{O}(d_F, d_B)$ and $\mathcal{O}(d_F, d_B)[1]$. Those which lie inside quiver domains must necessarily decay before reaching this domain. Initial rays coming from below the parabola are plotted in thick black lines.

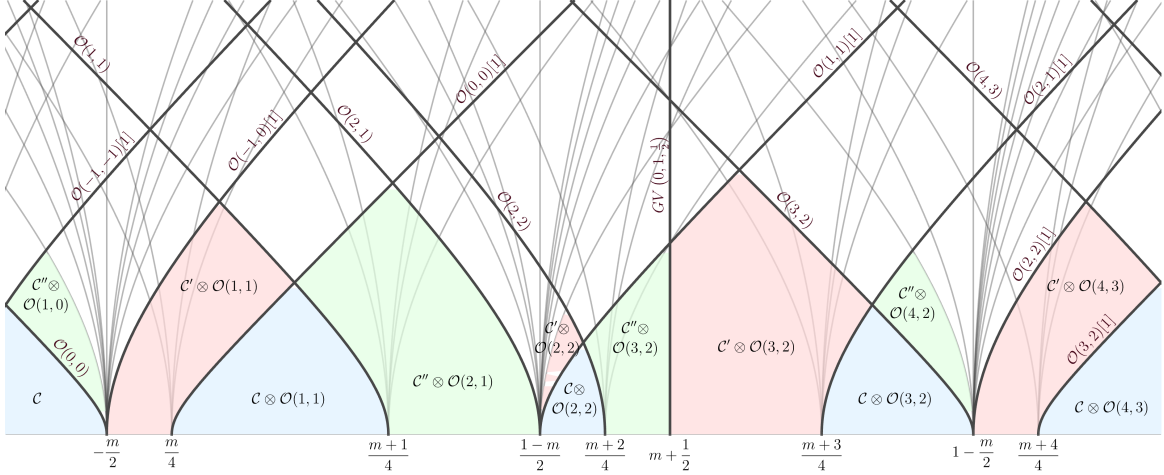


FIGURE 5. Similar as Figure 4, now assuming $0 < m < \frac{1}{3}$ (plotted for $m = \frac{1}{10}$).

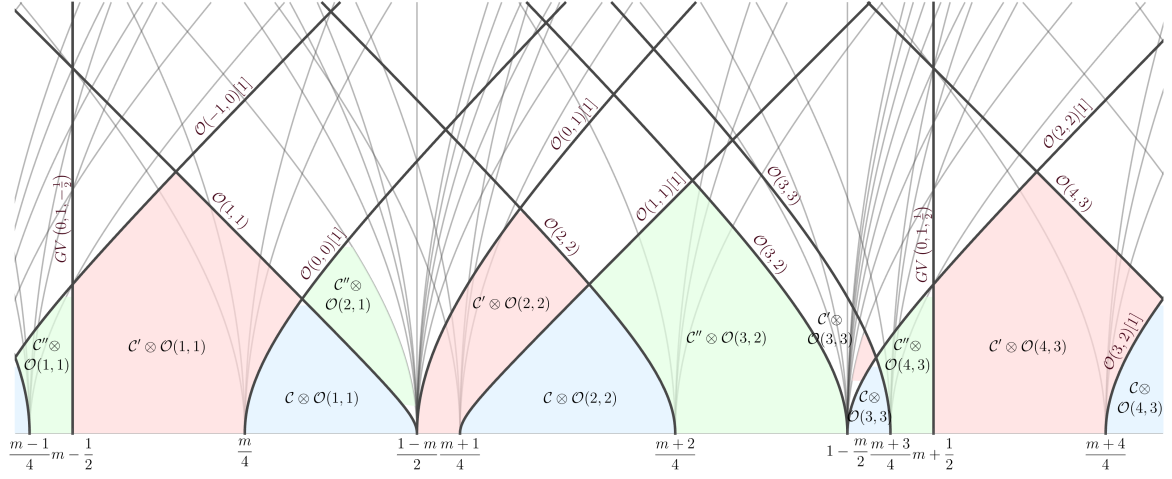


FIGURE 6. Similar as Figure 4, now assuming $\frac{1}{3} < m < \frac{2}{3}$ (plotted for $m = \frac{2}{5}$).

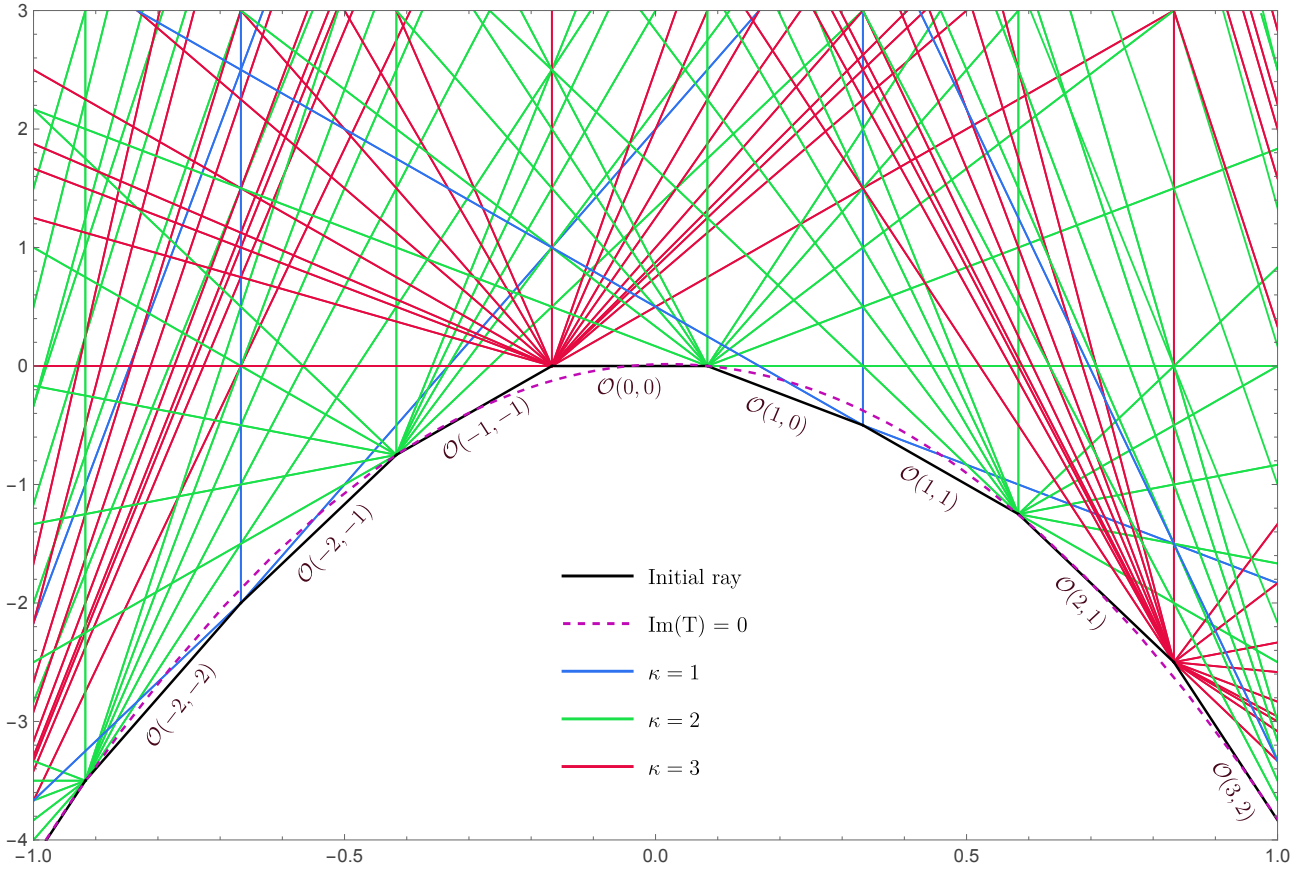


FIGURE 7. Initial and secondary rays in affine coordinates (x, y) for $-\frac{1}{3} < m < 0$ (here drawn for $m = -\frac{1}{6}$). Secondary rays originating from collisions with Dirac product $\kappa = 1, 2, 3$ are shown in blue, green, red, respectively. Collisions with $\kappa = 1, 2, 3$ occur below, at and above the parabola, respectively.

Charge (FB)	Charge (HC)	Contributing tree(s)	Index $\Omega_H(y)$ (nonneg. part)
$[0, 0, 1, \frac{1}{2}]$	$[0, 0, 1, \frac{1}{2}]$	$\{\mathcal{O}(1, 0)[1], \mathcal{O}(1, 1)\}$	1
$[0, 1, 0, 0]$	$[0, 1, -1, 0]$	$\{\mathcal{O}(0, 0)[1], \mathcal{O}(1, 0)\}$	$-y$
$[0, 1, 1, \frac{1}{2}]$	$[0, 1, 0, \frac{1}{2}]$	$\{\mathcal{O}(0, 0)[1], \mathcal{O}(1, 1)\}$	$y^2 + 1$
$[0, 2, 1, \frac{1}{2}]$	$[0, 2, -1, \frac{1}{2}]$	$\{\mathcal{O}(1, 1), \{2\mathcal{O}(0, 0)[1], \mathcal{O}(1, 0)\}\}$	$y^4 + y^2 + 1$
$[0, 2, 2, 0]$	$[0, 2, 0, 0]$	$\{\mathcal{O}(1, 1), \mathcal{O}(-1, -1)[1]\}$	$-y^5 - y^3 - y$
$[0, 3, 2, 0]$	$[0, 3, -1, 0]$	$\{\mathcal{O}(-1, -1)[1], \{\mathcal{O}(1, 1), \{\mathcal{O}(0, 0)[1], \mathcal{O}(1, 0)\}\}\}$	$-y^9 - 3y^7 - 4y^5 - 4y^3 - 4y$
$[0, 3, 3, \frac{1}{2}]$	$[0, 3, 0, \frac{1}{2}]$	$\{\mathcal{O}(-1, -1)[1], \{\mathcal{O}(0, 0)[1], 2\mathcal{O}(1, 1)\}\}$	$y^{10} + 2y^8 + 3y^6 + 3y^4 + 3y^2 + 3$
$[0, 4, 2, 0]$	$[0, 4, -2, 0]$	① $\{\mathcal{O}(-2, -1)[1], \mathcal{O}(2, 1)\}$ ② $\{\mathcal{O}(-1, -1)[1], \{\mathcal{O}(1, 1), \{2\mathcal{O}(0, 0)[1], 2\mathcal{O}(1, 0)\}\}\}$	$-y^{13} - 3y^{11} - 8y^9 - 10y^7 - 11y^5 - 11y^3 - 11y$
$[1, 2, 2, 2]$	$[1, 2, 0, 2]$	$\{\mathcal{O}(2, 1), \{\mathcal{O}(1, 0)[1], \mathcal{O}(1, 1)\}\}$	1
$[1, 0, 1, -\frac{1}{2}]$	$[1, 0, 1, -\frac{1}{2}]$	$\{\mathcal{O}(0, 0), \{\mathcal{O}(-2, -2)[1], \mathcal{O}(-2, -1)\}\}$	1
$[1, 1, 2, 0]$	$[1, 1, 1, 0]$	$\{\mathcal{O}(1, 1), \{\mathcal{O}(-2, -2)[1], \mathcal{O}(-2, -1)\}\}$	1
$[1, 2, 3, \frac{3}{2}]$	$[1, 2, 1, \frac{3}{2}]$	$\{\{\mathcal{O}(-2, -2)[1], \mathcal{O}(-2, -1)\}, \{\mathcal{O}(2, 1), \{\mathcal{O}(1, 0)[1], \mathcal{O}(1, 1)\}\}\}$	1
$-[1, 0, -1, -\frac{1}{2}]$	$-[1, 0, -1, -\frac{1}{2}]$	$\{\mathcal{O}(0, 0)[1], \{\mathcal{O}(1, 0)[1], \mathcal{O}(1, 1)\}\}$	1
$-[1, 2, 1, \frac{3}{2}]$	$-[1, 2, 1, \frac{3}{2}]$	$\{\mathcal{O}(2, 1)[1]\}$	1
$[1, 0, 0, -1]$	$[1, 0, 0, -1]$	① $\{\mathcal{O}(-2, -2)[1], 2\mathcal{O}(-1, -1)\}$ ② $\{\{\mathcal{O}(-2, -2)[1], \mathcal{O}(-2, -1)\}, \{\mathcal{O}(-2, -1)[1], 2\mathcal{O}(-1, -1)\}\}$	$y^2 + 2$
$[1, 0, 0, -2]$	$[1, 0, 0, -2]$	① $\{\mathcal{O}(-1, -1), \{\mathcal{O}(-3, -2)[1], \mathcal{O}(-2, -1)\}\}$ ② $\{\{\mathcal{O}(-5, -4)[1], \mathcal{O}(-5, -3)\}, \{\mathcal{O}(-2, -1)[1], 2\mathcal{O}(-1, -1)\}\}$ ③ $\{\{\mathcal{O}(-3, -2)[1], \mathcal{O}(-2, -2)\}, \{\mathcal{O}(-1, -1), \{\mathcal{O}(-2, -2)[1], \mathcal{O}(-2, -1)\}\}\}$	$y^4 + 3y^2 + 6$
$[2, 1, 1, -\frac{1}{2}]$	$[2, 1, 0, -\frac{1}{2}]$	$\{3\mathcal{O}(0, 0), \mathcal{O}(-1, -1)[1]\}$	1
$[2, 1, 0, -1]$	$[2, 1, -1, -1]$	$\{2\mathcal{O}(0, 0), \{\mathcal{O}(-2, -1)[1], \mathcal{O}(-1, -1)\}\}$	$-y$

TABLE 2. Charges, contributing trees for $m = -\frac{1}{10}$, and the nonnegative-power part of the refined index $\Omega_H(y)$ (the full Laurent polynomial is symmetric under $y \leftrightarrow y^{-1}$). The rank 0 entries are consistent with the refined D2-D0 indices in (2.12).

5. Π -STABILITY SLICE

5.1. **Mirror curve.** The canonical bundle over $\mathbb{F}_1$ is described by the following toric data,

$$(5.1) \quad \begin{array}{ccc|cc} 0 & 0 & 1 & -2 & -1 \\ 1 & 0 & 1 & 1 & 0 \\ -1 & -1 & 1 & 0 & 1 \\ -1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 & 1 \end{array}$$

The mirror curve in $\mathbb{C}^\times \times \mathbb{C}^\times$ then takes the form

$$(5.2) \quad H(e^x, e^p; z_1, z_2) = 1 + e^x + e^p + z_1 e^{-x} + z_1 z_2 e^{-x-p}.$$

After choosing coordinates $y_1 = e^x$, $y_2 = e^p$ and rescaling the equation we arrive at

$$(5.3) \quad H(y_{1,2}; z_{1,2}) = y_1^2 y_2 + y_1 y_2^2 + y_1 y_2 + z_1 \cdot y_2 + z_1 z_2 \cdot y_1^2.$$

This can be mapped to into the Weierstrass form

$$(5.4) \quad Y^2 = 4X^3 - g_2 X - g_3,$$

with coefficients given by

$$(5.5) \quad \begin{aligned} g_2 &= 108(1 - 8z_1 + 16z_1^2 + 24z_1 z_2), \\ g_3 &= -216(1 - 12z_1 + 48z_1^2 - 64z_1^3 + 36z_1 z_2 - 144z_1^2 z_2 + 216z_1^2 z_2^2), \end{aligned}$$

and the discriminant takes the form

$$(5.6) \quad \Delta = g_2^3 - 27g_3^2 = 2^{12} \cdot 3^{12} \cdot z_1^3 z_2^2 [(1 - 4z_1)^2 - z_2(1 - 36z_1 + 27z_1 z_2)]$$

such that the Klein invariant⁴ takes the form [12, (6.4)]

$$(5.7) \quad J(z_1, z_2) = 1728 \frac{g_2^3}{\Delta} = \frac{[(1 - 4z_1)^2 + 24z_1 z_2]^3}{z_1^3 z_2^2 [(1 - 4z_1)^2 - z_2(1 - 36z_1 + 27z_1 z_2)]}$$

The conifold locus $\mathcal{C} = \{(1 - 4z_1)^2 - z_2(1 - 36z_1 + 27z_1 z_2) = 0\}$ intersects $z_1 = 0$ at $z_2 = 1$ normally, and is tangent to $z_2 = 0$ at $z_1 = 1/4$. It will be useful to define

$$(5.8) \quad z_1 = \lambda u^2, \quad z_2 = -\frac{u}{\lambda} \quad \leftrightarrow \quad u = -(z_1 z_2)^{1/3}, \quad \lambda = \frac{z_1^{1/3}}{z_2^{2/3}}$$

and even better

$$(5.9) \quad u = u/\lambda = -z_2, \quad \ell = \lambda^3 = \frac{z_1}{z_2^2}$$

such that (5.7) becomes

$$(5.10) \quad J(u, \ell) = \frac{(1 - 8\ell u^2 - 24u^3 + 16\ell^2 u^4)^3}{\ell^3 u^8 (1 + u - 8\ell u^2 - 36\ell u^3 + \ell(16\ell - 27)u^4)}$$

$$(5.11) \quad = \frac{Q_4^3}{\ell^3 u^8 \Delta_4} = 1728 + \frac{Q_6^2}{\ell^3 u^8 \Delta_4}$$

where

$$(5.12) \quad \begin{aligned} \Delta_4 &= 1 + u - 8\ell u^2 - 36\ell u^3 + \ell(16\ell - 27)u^4 \\ Q_4 &= 1 - 8\ell u^2 - 24u^3 + 16\ell^2 u^4 \\ Q_6 &= 1 - 12u^2 \ell - 36u^3 \ell + 48u^4 \ell^2 + 144u^5 \ell^2 + u^6 \ell^2 (216 - 64\ell) \end{aligned}$$

⁴defined such that $J(\tau) = q^{-1} + 744 + 196884q + \dots$ such that $J(i) = 1728, J(e^{2\pi i/3}) = 0$.

For generic λ , the discriminant $\Delta = \ell^3 u^8 \Delta_4$ has 4 distinct roots at finite u and an order 8 root at $u = 0$, so the elliptic fibration is of Kodaira type $I_8, 4I_1$ in the notations of [32]. The equation $J(u, \ell) = j$ is polynomial of degree 12 in u , with discriminant proportional to

$$(5.13) \quad (27 + 256\ell)^3 j^8 (j - 1728)^6 [(27 + 256\ell)(243 + 256\ell)^3 - 2^{24} \ell^3 j]$$

Generically, u is a 12:1 covering⁵ of the j -line with ramification profile $1^4, 8$ at $j = \infty$, 3^4 at the $\mathbb{Z}_3$ orbifold point $j = 0$ (i.e. 4 pairs of multiplicity 3 roots, corresponding to the zeros of Q_4), 2^6 at the $\mathbb{Z}_2$ orbifold point $j = 1728$ (i.e. 6 pairs of multiplicity 2 roots, corresponding to the zeros of Q_6), and $1^{10}2$ at $j = j_B$ with

$$(5.14) \quad j_B = \frac{(27 + 256\ell)(243 + 256\ell)^3}{2^{24} \ell^3}$$

The corresponding double root is at $u = -\frac{8}{9}$, around which

$$(5.15) \quad J(u, \ell) = j_B - \frac{3^8(256\ell + 243)^2(65536\ell^2 - 124416\ell - 19683)}{2^{28}\ell^3(256\ell + 27)} \left(u + \frac{8}{9}\right)^2 + \mathcal{O}\left(u + \frac{8}{9}\right)^3$$

As we shall see later, the point $u = -\frac{8}{9}$ is an apparent singularity of the Picard-Fuchs equation, where the solutions are actually regular.

In contrast, for $\ell = -\frac{27}{256}$, the degree of the equation $J(u, \ell) = j$ drops to 10, with Δ_4, Q_4 and Q_6 all acquiring a common factor $9u + 8$ (of multiplicity 2 in the case of Δ_4):

$$(5.16) \quad \begin{aligned} \Delta_4 &= \frac{1}{4096} (9u + 8)^2 (153u^2 - 80u + 64) \\ Q_4 &= \frac{1}{4096} (9u + 8) (81u^3 + 1080u^2 - 576u + 512) \\ Q_6 &= \frac{1}{262144} (9u + 8) (72171u^5 - 17496u^4 + 31104u^3 + 82944u^2 - 36864u + 32768) \end{aligned}$$

In particular, the elliptic fibration is now of type $I_8, 2I_1, II$, with the fiber II corresponds to the Argyres-Douglas theory H_0 , with an elliptic point of order 3 [32, §5.2]. The u -line is now a 10:1 covering⁶ of the j -line with ramification profile $1^4, 8$ at $j = \infty$, $1, 3^3$ at $j = 0$ and 2^5 at $j = 1728$. We can now invoke [33, Theorem 1]⁷ to conclude that u is a Hauptmodul for a subgroup Γ of index 10 in $PSL(2, \mathbb{Z})$. This subgroup was identified in [16, Table 7] as the non-congruence subgroup $\Gamma(10; 0, 3, 0, 1; 8)$ of index 10 in $PSL(2, \mathbb{Z})$, with genus 0, 3 cusps of width $(8, 1, 1)$ and one $\mathbb{Z}_3$ orbifold point (using the notation of [34] for subgroups of $PSL(2, \mathbb{Z})$; note that the parameter λ in [32] corresponds to $\lambda^{3/2}$ in our conventions). By unfolding the standard fundamental domain $\mathcal{F}$ of $PSL(2, \mathbb{Z})$ into the fundamental domain $\mathcal{F}_\Gamma$ of Γ , we can associate a unique value $\underline{u}(\tau)$ among the ten solutions u_i with the same values of $J(u, \ell) = J(\tau)$, related to the standard Hauptmodul of Γ (see Appendix) by

$$(5.17) \quad \underline{u}(\tau) = \frac{8w}{J_\Gamma(\tau) + w}$$

where $w := -(-1/3^9)^{1/8} \simeq -0.268444 - 0.111193i$. The universal cover of the Kähler moduli space at $\ell = -27/256$ is therefore the Poincaré upper half-plane.

⁵Note that the ramification structure of the u -line over the j -line satisfies the Riemann-Hurwitz formula $\chi(S') = N\chi(S) - \sum_{P \in S'} (e_P - 1)$ for the Euler characteristic of an $N : 1$ cover $S' \rightarrow S$, where e_P is the ramification index at point P . Here, $\chi(S) = \chi(S') = 2 = 24 - (7 + 1) \times (2 - 1) - 4 \times (3 - 1) - 1 \times (8 - 1)$.

⁶Similar to footnote 5, it is easy to check that the ramification structure satisfies the Riemann-Hurwitz formula, $2 = 2 \times 10 - 5 \times (2 - 1) - 3 \times (3 - 1) - 1 \times (8 - 1)$. Equivalently, the genus of the congruence subgroup Γ is given by $g = 1 + \frac{\mu}{12} - \frac{|\mathcal{E}_2|}{4} - \frac{|\mathcal{E}_3|}{3} - \frac{|\mathcal{C}|}{2} = 1 + \frac{10}{12} - \frac{1}{3} - \frac{3}{2}$, where $|\mathcal{E}_2|$ is the number of elliptic points of order 2, $|\mathcal{E}_3|$ is the number of elliptic points of order 3, and $|\mathcal{C}|$ the number of cusps.

⁷Namely: A necessary and sufficient condition that $f(\tau)$ is a modular function on a subgroup of finite index in $PSL(2, \mathbb{Z})$ is that $f(\tau)$ should be an algebraic function of the Klein invariant J , and that its only branch points should be branch points of order *dividing* 2 at which $J = 1728$, branch points of order *dividing* 3 at which $J = 0$, and branch points at which J is infinite.

Away from the locus $\ell = \underline{\ell} := -\frac{27}{256}$, we can still attempt to parametrize the $\mathbf{u}$ -line by τ , by equating the Klein Hauptmodul at $\mathbf{u}(\tau, \ell)$ with its value at $\underline{\mathbf{u}}(\tau)$,

$$(5.18) \quad J(\tau) = \frac{Q_4^3}{\ell^3 \mathbf{u}^8 \Delta_4} = \frac{\underline{Q}_4^3}{\underline{\ell}^3 \underline{\mathbf{u}}^8 \underline{\Delta}_4}$$

In order to solve this equation, we can introduce an unknown non-vanishing function $f(\underline{\mathbf{u}}, \ell)$ and solve instead

$$(5.19) \quad Q_4 = f \underline{Q}_4, \quad Q_6 = f^{3/2} \underline{Q}_6, \quad \mathbf{u}^8 \ell^3 \Delta_4 = f^3 \underline{\mathbf{u}}^8 \underline{\ell}^3 \underline{\Delta}_4$$

where the last equation is a consequence of the first two. For $|\ell - \underline{\ell}|$ sufficiently small, ten of the roots of (5.18), which we denote by $\mathbf{u}_i$ ($i = 1, \dots, 10$) are small perturbations of $\underline{\mathbf{u}}_i$, which can be determined as a Taylor series in $\ell - \underline{\ell}$,

$$(5.20) \quad \begin{aligned} \mathbf{u} &= \underline{\mathbf{u}} \left(1 - \frac{4}{9}(\underline{\mathbf{u}} - 8)(\ell - \underline{\ell}) + \frac{16(27\underline{\mathbf{u}}^3 - 300\underline{\mathbf{u}}^2 + 2624\underline{\mathbf{u}} + 2816)}{243(9\underline{\mathbf{u}} + 8)}(\ell - \underline{\ell})^2 + \dots \right) \\ f &= 1 - \frac{16\underline{\mathbf{u}}^2}{9\underline{\mathbf{u}} + 8}(\ell - \underline{\ell}) + \frac{32\underline{\mathbf{u}}^2(45\underline{\mathbf{u}}^2 - 56\underline{\mathbf{u}} - 128)}{9(9\underline{\mathbf{u}} + 8)^2}(\ell - \underline{\ell})^2 + \dots \\ &= 1 - \frac{16\underline{\mathbf{u}}^2}{9\underline{\mathbf{u}} + 8}(\ell - \underline{\ell}) + \frac{32\underline{\mathbf{u}}^2(27\underline{\mathbf{u}}^2 + 56\underline{\mathbf{u}} + 128)}{9(9\underline{\mathbf{u}} + 8)^2}(\ell - \underline{\ell})^2 + \dots \end{aligned}$$

Note that this expansion blows up at $9\underline{\mathbf{u}} + 8 = 0$, reflecting the existence of a branch point at $\tau = \tau_B$ (and all its images under Γ), where τ_B is one of the 10 pre-images of $j = j_B$. The two additional roots, which we denote by $\mathbf{u}_{\pm}$, instead emerge from the ramification point, $\mathbf{u}_{\pm} = -\frac{8}{9} + \mathcal{O}(\ell - \underline{\ell})$. The expansion can be determined by matching the sum and product over all roots, which are easily determined from the coefficients in the degree 12 polynomial $Q_{12} = Q_4^3 - j\ell^3 \mathbf{u}^8 \Delta_4$,

$$(5.21) \quad \sum_{i=1}^{10} \underline{\mathbf{u}}_i = \frac{16(5j - 621)}{9(17j + 27)}, \quad \prod_{i=1}^{10} \underline{\mathbf{u}}_i = \frac{2^{30}}{3^{11}(17j + 27)}$$

$$(5.22) \quad \sum_{i=\pm,1}^{10} \mathbf{u}_i = \frac{36(j - 512\ell)}{j(16\ell - 27) - 4096\ell^2}, \quad \prod_{i=\pm,1}^{10} \mathbf{u}_i = \frac{1}{j(27 - 16\ell)\ell^4 + 4096\ell^6}$$

Using this and the expansion (5.20) for each $\mathbf{u}_i$, we find

$$(5.23) \quad \begin{aligned} \mathbf{u}_+ + \mathbf{u}_- &= -\frac{16}{9} - \frac{2048}{729}(\ell - \underline{\ell}) - \frac{1736704(\ell - \underline{\ell})^2}{59049} - \frac{1024458752(\ell - \underline{\ell})^3}{4782969} + \dots \\ \mathbf{u}_+ \mathbf{u}_- &= \frac{64}{81} + \frac{16384}{6561}(\ell - \underline{\ell}) + \frac{4980736(\ell - \underline{\ell})^2}{177147} + \frac{1098907648(\ell - \underline{\ell})^3}{4782969} + \dots \end{aligned}$$

In particular the dependence on j drops, and the discriminant $(\mathbf{u}_+ - \mathbf{u}_-)^2$ vanishes up to order $\mathcal{O}(\ell - \underline{\ell})^3$. As a result, we find

$$(5.24) \quad \mathbf{u}_{\pm} = -\frac{8}{9} - \frac{1024}{729}(\ell - \underline{\ell}) \pm \frac{4096\sqrt{5}(\ell - \underline{\ell})^{3/2}}{6561} - \frac{868352(\ell - \underline{\ell})^2}{59049} + \mathcal{O}(\ell - \underline{\ell})^{5/2}$$

Beware: in this computation, I assumed that $\mathbf{u}_i$ was a polynomial in $\mathbf{u}_i$, and used the fact that for any pair of polynomials $P(z), Q(z)$,

$$(5.25) \quad \sum_{i: P(z_i)=0} Q(z_i) = \oint_{\infty} Q(z) \frac{P'(z)}{P(z)} \frac{dz}{2\pi iz}$$

Here the relevant $Q(z)$ is instead a rational function of $\underline{\mathbf{u}}$ with a pole at $\underline{\mathbf{u}} = -\frac{8}{9}$, at least in a series expansion around $\ell = \underline{\ell}$. Upon deforming the contour from circles around z_i to ∞ , one therefore picks up an additional contribution from this pole. It turns out that for $\underline{\mathbf{u}} = -\frac{8}{9}$, the ratio $P'(z)/P(z)$ evaluates to $-\frac{83}{3} + \frac{27648}{j}$. Subtracting this contribution will therefore introduce

j dependence at each order in $(\ell - \underline{\ell})$. It is unclear however if the limits $\ell \rightarrow \underline{\ell}$ and $z \rightarrow \infty$ commute.

Finally, for $\lambda = 0$, one of the I_1 fibers collide with the I_8 fiber at $u = 0$, leading to a configuration of type $I_9, 3I_1$. This corresponds to the blow-down limit where the exceptional curve C shrinks to a point (see below). The Coulomb branch is now a modular curve for the congruence subgroup $\Gamma_0(9)$, which has index 12 in $SL(2, \mathbb{Z})$ and 4 cusps of width $(9, 1, 1, 1)$. Keeping $z = z_1 z_2 = -u^3$ fixed and taking the limit $z_1 \rightarrow 0$, the discriminant and J-invariant reduce to

$$(5.26) \quad \Delta \rightarrow -z^3(1 + 27z), \quad J \rightarrow -\frac{(1 + 24z)^3}{z^3(1 + 27z)}$$

Equating

$$(5.27) \quad J(\tau) = \frac{(J_3 + 15)(J_3 + 231)^3}{(J_3 - 12)^3}, \quad J_3 = \frac{(J_9 - 3)^3}{J_9^2 + 3J_9 + 9} + 12$$

we can express u in terms of J_9 order by order in λ ,

$$(5.28) \quad 1/u = \frac{3(J_9 + 6)}{J_9 - 3} \left(1 + \frac{(J_9 + 6)\lambda}{3(J_9 - 3)} - \frac{(J_9^3 + 72J_9^2 + 270J_9 + 702)\lambda^2}{9(J_9 - 3)^2(J_9 + 6)} + \dots \right)$$

In terms of the Fricke transform $\hat{J}_9 = \frac{27}{J_9 - 3} + 3$,

$$(5.29) \quad 1/u = \hat{J}_9 \left(1 + \frac{\hat{J}_9 \lambda}{9} + \frac{(18 - \hat{J}_9^3)\lambda^2}{27\hat{J}_9} + \frac{7(5\hat{J}_9^3 - 108)\lambda^3}{2187} - \frac{11\hat{J}_9(13\hat{J}_9^3 - 297)\lambda^4}{19683} + \dots \right)$$

I haven't managed to identify the rational function resumming up this expansion.

We note two further special cases:

- For $\ell^3 = \frac{27}{16}$, one of the roots of Δ_4 moves off to $u = \infty$, but this does not affect the structure of the singular fibers.
- In the limit $\lambda \rightarrow \infty$, the 4 roots of Δ_4 split into two pairs

$$(5.30) \quad 1/u_{\pm} = -2\sqrt{\lambda} \left(1 \pm \frac{i}{\lambda^{3/4}} + \frac{1}{8\lambda^{3/2}} \mp \frac{i}{32\lambda^{9/4}} + \dots \right),$$

$$(5.31) \quad 1/u'_{\pm} = 2\sqrt{\lambda} \left(1 \pm \frac{1}{\lambda^{3/4}} - \frac{1}{8\lambda^{3/2}} \pm \frac{1}{32\lambda^{9/4}} + \dots \right)$$

Setting $1/u = -2\sqrt{\lambda} \left(1 + \frac{i u_{\text{SW}}}{2\lambda^{3/4}} + \mathcal{O}(\lambda^{-3/2}) \right)$ or $1/u = 2\sqrt{\lambda} \left(1 + \frac{u_{\text{SW}}}{2\lambda^{3/4}} + \mathcal{O}(\lambda^{-3/2}) \right)$ and taking the limit $\lambda \rightarrow \infty$ with u_{SW} fixed⁸, we recover in either case the usual Klein invariant for $SU(2)$ Seiberg-Witten theory in 4 dimensions,

$$(5.32) \quad J = \frac{256(u_{\text{SW}}^2 - 3)^3}{u_{\text{SW}}^2 - 4} + \mathcal{O}(\lambda^{-3/4})$$

Out of place: To match the notations in [32], we define (note a missing term $+144U\lambda_{CM}$ in the formula for $g_3(U)$ in (5.85), loc.cit.)

$$(5.33) \quad U = \frac{1}{\sqrt{z_1}}, \quad \lambda_{CM} = -\frac{z_2}{\sqrt{z_1}} = -\lambda^{-3/2}$$

such that the mirror curve [32, (2.94)] becomes

$$(5.34) \quad \sqrt{\lambda_{CM}} \left(\frac{1}{t} + tw \right) + w + \frac{1}{w} - U = 0$$

⁸This is equivalent to the prescription $z_1 \rightarrow \frac{1}{4} \exp(-2\sqrt{2}\epsilon^2 \tilde{u})$, $z_2 \sim \epsilon^4 \Lambda^4$ from [13, 6.33].

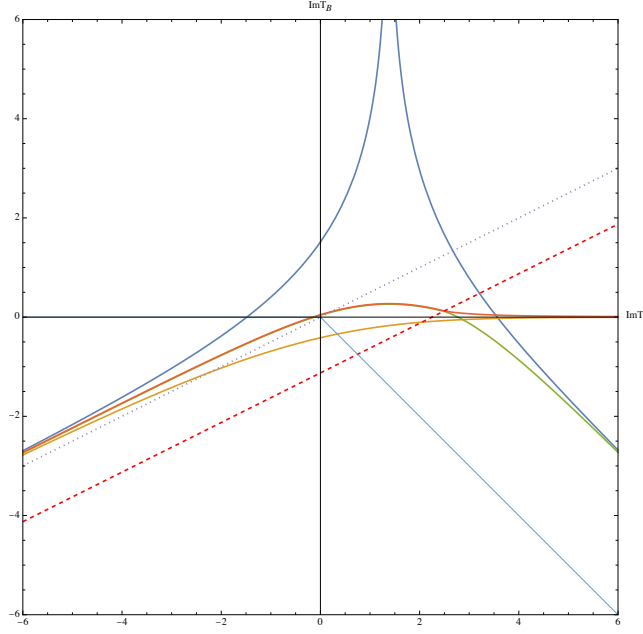


FIGURE 8. Real slice of the discriminant locus for local $\mathbb{F}_1$. The slices $|\lambda| = cte$ correspond to lines of fixed $\Im(T_F - 2T_B)$. The Kähler cone of $\mathbb{F}_1$ corresponds to the first quadrant $\Im T_F > 0, \Im T_B > 0$. The Kähler cone of the flopped geometry corresponds to $\Im T_B < 0, \Im(T_F + T_B) > 0$ i.e. $\Im T_C > 0, \Im T_H > 0$. The dashed and dotted lines correspond to loci with fixed λ equal to $(-27/256)^{1/3}$ and 1, respectively.

The mirror curve of local $\mathbb{P}^2$

$$(5.35) \quad \frac{1}{t} + \frac{1}{w} + tw - 3U = 0$$

is recovered upon redefining $(t, w, U) \rightarrow (\lambda_{CM}^{1/6}t, w/\lambda_{CM}^{1/3}, 3U\lambda_{CM}^{1/3})$ and taking the limit $\lambda_{CM} \rightarrow \infty$ keeping z fixed. Equivalently, once can set $z_2 = z/z_1$ and take the limit $z_1 \rightarrow 0$.

5.2. Large volume expansion. The Picard-Fuchs system is given by [12] (after fixing a typo in loc. cit.)

$$(5.36) \quad \begin{aligned} \mathcal{D}_1 &= \theta_1(\theta_1 - \theta_2) - z_1(2\theta_1 + \theta_2)(1 + 2\theta_1 + \theta_2) \\ \mathcal{D}_2 &= \theta_2^2 - z_2(\theta_2 - \theta_1)(2\theta_1 + \theta_2) \end{aligned}$$

In terms of the coordinates $(\tilde{u} = -1/u, \lambda)$ defined in (5.8),

$$(5.37) \quad \begin{aligned} 3\mathcal{D}_1 &= -3u^4\lambda\partial_u^2 - 6u^3\lambda\partial_u + u\partial_u\lambda\partial_\lambda + \lambda\partial_\lambda f + \lambda^2\partial_{\lambda^2} \\ 9\mathcal{D}_2 &= 4\lambda^2\partial_\lambda^2 + 4\lambda\partial_\lambda + u\partial_u - (4\lambda + 9u)u\partial_u\partial_\lambda + u^2\partial_u^2 \end{aligned}$$

Combining the two equations one arrives at the third order differential operator

$$(5.38) \quad \begin{aligned} \mathcal{D}_3 &= (8\lambda + 9u) \left(\lambda + (16\lambda^3 - 27)u^4 - 36\lambda u^3 - 8\lambda^2 u^2 + u \right) \theta_u^3 \\ &\quad - u \left(\lambda + (729 - 432\lambda^3)u^4 + 8\lambda(189 - 64\lambda^3)u^3 + 936\lambda^2 u^2 + 128\lambda^3 u \right) \theta_u^2 \\ &\quad + 2u^2 \left(-32\lambda^3 + 9(16\lambda^3 - 27)u^3 + 6\lambda(32\lambda^3 - 81)u^2 - 282\lambda^2 u \right) \theta_u \\ \lambda\mathcal{D}_3 &= (9u + 8) \left(16u^4\ell^2 - (9u(3u + 4) + 8)u^2\ell + u + 1 \right) \theta_u^3 \\ &\quad + u \left(16(27u + 32)u^3\ell^2 - (729u^3 + 1512u^2 + 936u + 128)u\ell - 1 \right) \theta_u^2 \\ &\quad + 2u^2\ell \left(9u^3(16\ell - 27) + 6u^2(32\ell - 81) - 282u - 32 \right) \theta_u \end{aligned}$$

with $\theta_u = u\partial_u$, $\theta_u = u\partial_u$, along with

$$(5.39) \quad \mathcal{D}_4 = (8\lambda + 9u)\partial_u\partial_\lambda f - (24\lambda u^2 + 1)\partial_u f - u(12\lambda u^2 + 1)\partial_u^2 f$$

One may check that (5.38) agrees with [13, (6.28)] upon renaming $(\tilde{u}, m)$ into $(-1/u, \lambda)$ and reordering the derivatives.

Importantly, the coefficient of θ_u^3 is proportional to $(9u + 8)\Delta_4$, hence vanishes on the finite discriminant locus as well as for $u = -8/9$. For $\ell \neq \underline{\ell}$, the latter coincides with the ramification point (5.15) and corresponds to an apparent singularity with trivial monodromy. Indeed a local analysis shows that the characteristic exponents at that point are 0, 1, 3, compared to 0, 1, 2 at an ordinary regular point. For $\ell = \underline{\ell}$, it coincides instead with the $\mathbb{Z}_3$ orbifold point of Γ and corresponds to an irregular point with characteristic exponents $(0, \frac{5}{6}, \frac{7}{6})$.

The operators $\mathcal{D}_3$ and $\mathcal{D}_4$ annihilate constants as well as $\log \lambda = \frac{1}{3} \log(z_1/z_2^2)$, so $\varpi_0 = 1$ and $\varpi_1 = \log z_1 - 2 \log z_2 = 3 \log \lambda$. The other logarithmic and double logarithmic solutions have expansions at small u given by

$$\begin{aligned} \varpi_2 &= \log u + u^2 \lambda + 2u^3 + \frac{3}{2} \lambda^2 u^4 + 12 \lambda u^5 + \dots \\ \varpi_3 &= (\log u)^2 - \varpi_2(2 \log u + \frac{1}{4} \log \lambda) + \frac{1}{8} (\log \lambda)^2 - \frac{u}{4\lambda} + \frac{1 - 16\lambda^3}{16\lambda^2} u^2 - \frac{90\lambda^3 + 1}{36\lambda^3} u^3 \\ &\quad + \frac{-208\lambda^6 - 24\lambda^3 + 1}{64\lambda^4} u^4 + \frac{6450\lambda^6 - 50\lambda^3 + 3}{300\lambda^5} u^5 + \dots \end{aligned} \quad (5.40)$$

We can in fact obtain an all-order solution as follows. First, observe that the operator (5.38) factorizes as $\mathcal{D}_3 = \mathcal{D}\theta_u$ where $\mathcal{D}$ is a second order differential operator. This operator must annihilate the periods ω_1, ω_2 of the elliptic curve. The fundamental period ω_1 is easily found from the GLSM charges,

$$(5.41) \quad \omega_1 = \sum_{k, \ell \geq 0} \frac{(k+2\ell)!}{(k!)^2 \ell! (\ell-k)!} u^{k+2\ell} \lambda^{\ell-k} = \sum_{k, \ell \geq 0} \frac{(3k+2\ell)!}{(k!)^2 \ell! (k+\ell)!} u^{3k+2\ell} \lambda^\ell$$

where in the second expression we shifted $\ell \mapsto \ell + k$, observing that terms with $k > \ell$ do not contribute in the sum in the first line. Using the Frobenius method (i.e. shifting $k \rightarrow k + x$, or $\ell \rightarrow \ell + y$, and taking the coefficient in the linear term in x or y), we find two logarithmic solutions **Fixed formulae below:**

$$\begin{aligned} \omega_2 &= (\log u - \log \lambda) \omega_1 + 2 \sum_{-2\ell < k \leq \ell} \frac{(k+2\ell)! (H_{2\ell+k} + H_{\ell-k} - 2H_k)}{(k!)^2 (k)! (\ell-k)!} u^{k+2\ell} \lambda^{\ell-k} \\ &\quad + \sum_{0 \leq \ell < k} (-1)^{k+\ell} \frac{\Gamma(k-\ell) (k+2\ell)!}{\ell! (k!)^2} u^{k+2\ell} \lambda^{\ell-k} \\ \omega'_2 &= (2 \log u + \log \lambda) \omega_1 + 2 \sum_{-2\ell < k \leq \ell} \frac{(k+2\ell)! (2H_{2\ell+k} - H_{\ell-k} - H_\ell)}{(k!)^2 (k)! (\ell-k)!} u^{k+2\ell} \lambda^{\ell-k} \\ &\quad - \sum_{0 \leq \ell < k} (-1)^{k+\ell} \frac{\Gamma(k-\ell) (k+2\ell)!}{\ell! (k!)^2} u^{k+2\ell} \lambda^{\ell-k} \end{aligned} \quad (5.42)$$

where $H_n = \sum_{k=1}^n (1/k)$ are the harmonic numbers. In fact, only the linear combination $2\omega_2 + 3\omega'_2$ is annihilated by $\mathcal{D}$, while ω_2 and ω'_2 satisfy it with a source term proportional to $9\lambda/u^4 + 96\lambda^3/u^3$. In particular, we find that the modular parameter is given by

$$(5.43) \quad \frac{\tau}{8} = -\frac{2\omega_2 + 3\omega'_2}{4\pi i \omega_1} = \log u + \frac{1}{8} \log \lambda + \frac{u}{8\lambda} + \frac{(32\lambda^3 - 1)u^2}{16\lambda^2} + \frac{(132\lambda^3 + 1)u^3}{24\lambda^3} + \dots$$

The CY periods satisfy $u\partial_u\varpi_2 = \omega_1, u\partial_u\varpi_3 = \frac{1}{4}(2\omega_2 + 3\omega'_2)$. Hence they are obtained by integrating once with respect to u ,

(5.44)

$$\begin{aligned}\varpi_2 &= \log u + \sum_{3k+2\ell>0} \frac{(3k+2\ell-1)!}{(k!)^2(\ell!)(k+\ell)!} u^{3k+2\ell} \lambda^\ell \\ \varpi_3 &= \log^2 u - (2\log u + \frac{1}{4}\log \lambda)\varpi_2 + \frac{1}{8}\log^2 \lambda \\ &\quad - 2 \sum_{-2\ell < k \leq \ell} \frac{(k+2\ell-1)!(2H_{2\ell+k} - H_{\ell-k} - H_\ell)}{(k!)^2(k)!(\ell-k)!} u^{k+2\ell} \lambda^{\ell-k} \\ &\quad + \sum_{0 \leq \ell < k} (-1)^{k+\ell} \frac{\Gamma(k-\ell)(k+2\ell-1)!}{\ell!(k!)^2} u^{k+2\ell} \lambda^{\ell-k} - 2 \sum_{k+2\ell>0} \frac{(k+2\ell-1)!}{(k+2\ell)(k!)^2\ell!(\ell-k)!} u^{k+2\ell} \lambda^{\ell-k}\end{aligned}$$

At any rate, in the large volume limit $u \rightarrow 0$, we get

$$\begin{aligned}(5.45) \quad \varpi_2 &\rightarrow \log u = \frac{1}{3}\log z_1 + \frac{1}{3}\log z_2 \\ \varpi_3 &\rightarrow -(\log u)^2 - \frac{1}{4}\log u \log \lambda + \frac{1}{8}(\log \lambda)^2 = -\frac{1}{8}(\log z_1)^2 - \frac{1}{4}\log z_1 \log z_2\end{aligned}$$

In order to match the desired large volume limit, we define

$$(5.46) \quad 2\pi i T_1 = 2\varpi_2 + \log \lambda, \quad 2\pi i T_2 = \varpi_2 - \log \lambda, \quad T_D = -4\frac{\varpi_3}{(2\pi i)^2} + \frac{1}{6}$$

Setting $T = \frac{1}{3}(T_1 + T_2) = \varpi_2/(2\pi i)$, $Q = e^{2\pi i T}$ and inverting the series, we get

$$(5.47) \quad -u = Q - \lambda Q^3 + 2Q^4 + \lambda^2 Q^5 - (\lambda^3 + 1)Q^7 + 4\lambda^2 Q^8 + \dots$$

and therefore **Recheck signs**

(5.48)

$$\varpi_3 = -(\log Q)^2 - \frac{1}{4}\log Q \log \lambda + \frac{1}{8}(\log \lambda)^2 + \frac{Q}{4\lambda} + \frac{1-16\lambda^3}{16\lambda^2}Q^2 + \frac{81\lambda^3+1}{36\lambda^3}Q^3 + \frac{1-16\lambda^6}{64\lambda^4}Q^4 + \dots$$

Integrating T_D/Q with respect to Q to get the free energy and substituting $Q = (Q_1 Q_2)^{1/3}$, $\lambda = (Q_1/Q_2)^{1/3}$, we recover the known GV invariants,

$$\begin{aligned}(5.49) \quad F &= -\frac{1}{6} \left(-3\log(\lambda)\log^2(Q) + 3\log^2(\lambda)\log(Q) + \frac{1}{6}\log Q - 8\log^3(Q) \right) \\ &\quad - \frac{Q}{\lambda} - \frac{(1-16\lambda^3)Q^2}{8\lambda^2} - \frac{(81\lambda^3+1)Q^3}{27\lambda^3} - \frac{(1-16\lambda^6)Q^4}{64\lambda^4} - \frac{(625\lambda^6+1)Q^5}{125\lambda^5} \\ &\quad - \frac{(-16\lambda^9-1215\lambda^6+1)Q^6}{216\lambda^6} - \frac{(2401\lambda^9+1)Q^7}{343\lambda^7} + \frac{(16\lambda^{12}+16384\lambda^9-1)Q^8}{512\lambda^8} + \dots \\ &= \frac{1}{162}(T_1 + T_2)(8T_1^2 + 25T_1T_2 - 10T_2^2) - (\text{Li}_3(Q_2) - 2\text{Li}_3(Q_1) + 3\text{Li}_3(Q_1Q_2) \\ &\quad + 5\text{Li}_3(Q_1^2Q_2) - 6\text{Li}_3(Q_1^2Q_2^2) + 7\text{Li}_3(Q_1^3Q_2) - 32\text{Li}_3(Q_1^3Q_2^2) + 27\text{Li}_3(Q_1^3Q_2^3) + \dots)\end{aligned}$$

Note in particular that the diagonal invariants $n_{d,d}^{(0)} = 3, -6, 27, \dots$ reproduce the genus 0 GV invariants of local $\mathbb{P}^2$. After the flop transition, using $\text{Li}_3(e^{-2\pi i x}) = \text{Li}_3(e^{2\pi i x}) + \frac{(2\pi i)}{6}(x^3 - \frac{3x^2}{2} + \frac{x}{2})$ we find that the polynomial term in F becomes

$$(5.50) \quad F' = \frac{16T_1'^3 - 18T_1'^2T_2' - 54T_1'T_2'^2 - 54T_2'^3 + 18T_1' + 81T_2'^2 - 27T_2'}{324} + \mathcal{O}(Q')$$

We can now express the periods in terms of the modulus τ by expanding the J -invariant in (5.7) at small u , and inverting the q -expansion:

$$(5.51) \quad 2\pi i\tau = 8 \log u + \log \lambda + \frac{u}{\lambda} + \frac{32\lambda^3 - 1}{2\lambda^2} u^2 + \frac{132\lambda^3 + 1}{3\lambda^3} u^3 + \frac{128\lambda^6 + 4\lambda^3 - 1}{4\lambda^4} u^4 + \dots$$

$$(5.52) \quad q_8 := e^{2\pi i\tau/8} = \lambda^{1/8} u + \frac{u^2}{8\lambda^{7/8}} - \frac{(256\lambda^3 - 7) u^3}{128\lambda^{15/8}} + \frac{(5888\lambda^3 + 35) u^4}{1024\lambda^{23/8}} + \dots$$

leading to

$$(5.53) \quad u^{-1} = \frac{\lambda^{1/8}}{q_8} + \frac{1}{8\lambda} + \frac{256\lambda^3 - 9}{128\lambda^{17/8}} q_8 + \frac{5(256\lambda^3 + 3)}{256\lambda^{13/4}} q_8^2 + \frac{(256\lambda^3 + 11)(256\lambda^3 + 171)}{32768\lambda^{35/8}} q_8^3 + \dots$$

Plugging into the large volume expansions (??), we find **Fixed signs**

$$(5.54) \quad \begin{aligned} 2\pi i T &= \log\left(\frac{q_8}{\lambda^{1/8}}\right) - \frac{q_8}{8\lambda^{9/8}} + \frac{5 - 64\lambda^3}{64\lambda^{9/4}} q_8^2 - \frac{9216\lambda^3 + 209}{3072\lambda^{27/8}} q_8^3 + \dots \\ (2\pi i)^2 T_D &= \frac{-27 \log^2 \lambda + 192 \log^2 q_8}{48} + \frac{(2\pi i)^2}{6} - \frac{\log q_8 - 1}{\lambda^{9/8}} q_8 - \frac{(64\lambda^3 - 5)(2 \log q_8 - 1)}{16\lambda^{9/4}} q_8^2 \\ &\quad - \frac{(9216\lambda^3 + 209)(3 \log q_8 - 1)}{1152\lambda^{27/8}} q_8^3 + \frac{(768\lambda^6 + 672\lambda^3 + 35)(4 \log q_8 - 1)}{256\lambda^{9/2}} q_8^4 + \dots \end{aligned}$$

In order to express this as an Eichler type integral, we recall that the algebraic Yukawa couplings are [12, (6.9)] (Note the likely sign error in front of coefficient 60)

$$(5.55) \quad \begin{aligned} C_{111} &= \frac{-1 - 4z_1^2 + z_2 - z_1(7 - 6z_2)}{3\Delta' z_1^3}, \quad C_{112} = \frac{-1 + 8z_1^2 + z_2 + z_1(2 - 3z_2)}{3\Delta' z_1^2 z_2}, \\ C_{122} &= \frac{z_2(1 - 12z_1) - (1 - 4z_1)^2}{3\Delta' z_1 z_2^2}, \quad C_{222} = \frac{2(1 - 4z_1)^2 + z_2(1 + 60z_1)}{3\Delta' z_2^3} \end{aligned}$$

Changing coordinates, one finds

$$(5.56) \quad C_{uuu} = C_{111}(\partial_u z_1)^3 + 3C_{112}(\partial_u z_1)^2 \partial_u z_2 + 3C_{122} \partial_u z_1 (\partial_u z_2)^2 + C_{222}(\partial_u z_2)^3 = -\frac{9u + 8\lambda}{u^3 \Delta'}$$

The holomorphic Yukawa coupling, normalized to agree with $\partial_T^3 F$ (up to powers of $2\pi i$), is then

$$(5.57) \quad C_{TTT} = C_{uuu} / (\partial_u \varpi_2)^3$$

Defining

$$(5.58) \quad \begin{aligned} C &:= \frac{8}{C_{TTT}} := \sum_{n \geq 0} c_n(\lambda) q_8^n \\ &= 1 - \frac{q_8}{8\lambda^{9/8}} + \frac{(5 - 64\lambda^3) q_8^2}{32\lambda^{9/4}} - \frac{209 + 9216\lambda^3}{1024\lambda^{27/8}} q_8^3 + \frac{(768\lambda^6 + 672\lambda^3 + 35) q_8^4}{128\lambda^{9/2}} + \dots \end{aligned}$$

we get

$$(5.59) \quad 2\pi i T = \log q_8 - \frac{1}{8} \log \lambda + \sum_{n \geq 1} \frac{c_n(\lambda)}{n} q_8^n$$

$$(5.60) \quad (2\pi i)^2 T_D = 4(\log q_8)^2 - \frac{9}{16}(\log \lambda)^2 + \frac{(2\pi i)^2}{6} + 8 \log q_8 \sum_{n \geq 1} \frac{c_n(\lambda)}{n} q_8^n - 8 \sum_{n \geq 1} \frac{c_n(\lambda)}{n^2} q_8^n$$

such that $\partial T_D / \partial T = \frac{8 \log q_8}{2\pi i} = \tau$. Equivalently,

$$(5.61) \quad \boxed{\begin{pmatrix} T \\ T_D \end{pmatrix} = \text{cte} + \int^\tau \begin{pmatrix} 1 \\ \rho \end{pmatrix} C(\rho) d\rho}$$

To obtain an explicit expression for C , we introduce $\omega_1 = u\partial_u\varpi_2$ and note that

$$(5.62) \quad \frac{C}{\omega_1} = u^{-1}\partial_\tau u, \quad \frac{C}{(\omega_1)^3} = \frac{1}{u^3 C_{uuu}} = -\frac{\Delta'}{9u + 8\lambda}$$

which allows to compute C and ω_1 separately, up to sign:

$$(5.63) \quad C^2 = -\left(u^{-1}\partial_\tau u\right) \frac{9u + 8\lambda}{\Delta'}, \quad \omega_1^2 = -\left(u^{-1}\partial_\tau u\right)^3 \frac{9u + 8\lambda}{\Delta'}$$

For $\lambda = \underline{\lambda}$, we find that C^2 is a holomorphic modular form of weight 6 under the non-congruence subgroup Γ , with a simple zero at the $\mathbb{Z}_3$ orbifold point double zeros at the conifold points

$$(5.64) \quad C^2 = \frac{(q_8 \partial_{q_8} J_\Gamma)^3}{(J_\Gamma + 10w)(J_\Gamma^2 - 8wJ_\Gamma + 144w^2)},$$

while $(\omega_1)^2$ is a meromorphic modular form of weight 2, with a simple pole at the orbifold point and a double zero at $u = \infty$,

$$(5.65) \quad \omega_1^2 = \frac{(J_\Gamma + w)^2 q_8 \partial_{q_8} J_\Gamma}{(J_\Gamma + 10w)(J_\Gamma^2 - 8wJ_\Gamma + 144w^2)}$$

The ratio C/ω_1^3 is modular invariant, vanishes at the orbifold and conifold points but has a triple pole at $u = \infty$,

$$(5.66) \quad \frac{C}{\omega_1^3} = -\frac{(J_\Gamma + 10w)(J_\Gamma^2 - 8wJ_\Gamma + 144w^2)}{(J_\Gamma + w)^3} = -\frac{1}{54}(8u + 8\underline{\lambda})(153u^2 - 80u\underline{\lambda} + 64\underline{\lambda}^2)$$

For general values of λ , we can compute C by deriving the Klein Hauptmodul (5.18) with respect to u and $\underline{u}$,

$$(5.67) \quad \partial_u J = -\frac{(9u + 8)Q_4^2 Q_6}{u^9 \ell^3 \Delta_4^2}, \quad \partial_{\underline{u}} J = -\frac{(9\underline{u} + 8)\underline{Q}_4^2 \underline{Q}_6}{\underline{u}^9 \underline{\ell}^3 \underline{\Delta}_4^2},$$

giving access to the derivative

$$(5.68) \quad \frac{\partial \underline{u}}{\partial u} = \frac{(9\underline{u} + 8)\underline{u}^7 \underline{\ell}^3}{(9u + 8)u^7 \ell^3} f^{5/2}$$

hence to the ratio

$$(5.69) \quad \left(\frac{C}{\underline{C}}\right)^2 = \frac{(9\underline{u} + 8)^2 \underline{u}^{16} \underline{\ell}^6}{(9u + 8)^2 u^{16} \ell^6} f^{9/2}$$

Note that f can only vanish when Q_4 and Q_6 admit a common root, which only occurs when their resultant vanishes, i.e. when $\ell = \underline{\ell}$ (but then $f = 1$) or $\ell = 0$ (but then $Q_4 = Q_6 = 1$, unless one rescales u).

5.3. Expansion near conifold points. Here, we analyze the behavior of the periods T and T_D near points u_c where $\Delta(u, \ell) = 0$. As mentioned in the introduction, it is advantageous to parametrize ℓ in terms of a new complex variable b , such that both ℓ and u_c are rational functions in b :

$$(5.70) \quad \ell = -\frac{b(b^2 + b + 1)^3}{(b + 1)^8}, \quad u_c = -\frac{(b + 1)^4}{(b^2 + b + 1)(b^2 + 4b + 1)}$$

For given ℓ , there are in general 8 solutions for b , which appear in pairs $(b, 1/b)$. The b -plane is therefore an 8:1 cover of the ℓ -plane, ramified at $\ell \in \{-27/256, 0, \infty\}$. In fact, it is possible to choose a simpler parametrization in terms of $\beta = b + 1/b$,

$$(5.71) \quad \ell = -\frac{(\beta + 1)^3}{(\beta + 2)^4}, \quad u_c = -\frac{(\beta + 2)^2}{(\beta + 1)(\beta + 5)}$$

where there are now in general 4 values of β for a given ℓ , which correspond to the four roots of the polynomial Δ_4 . However, it will turn out that the periods are not invariant under $b \rightarrow 1/b$, so we prefer to use the variable b throughout.

Because the characteristic exponents at $u = u_c$ of the third-order Picard-Fuchs equation $\mathcal{D}_3\varpi = 0$ are $(0, 1, 1)$, the solutions to the s $\mathcal{D}_1\varpi = \mathcal{D}_2\varpi = 0$ behave either as $\mathcal{O}(\delta)$, where $\delta = u - u_c$, or as $\mathcal{O}(\delta \log \delta)$. In the first case, plugging in the ansatz

$$(5.72) \quad \varpi_2^c = A_1(b)\delta + A_2(b)\delta^2 + A_3(b)\delta^3 + \dots$$

the Picard-Fuchs equation $\mathcal{D}_1\varpi = 0$ implies that $A_1(b)$ satisfies the differential equation

$$(5.73) \quad (b^6 + 5b^5 + 5b^4 - 5b^2 - 5b - 1) \partial_b A_1 + (5b^4 + 4b^3 + 18b^2 + 4b + 5) A_1(b) = 0$$

which is solved by (up to an overall multiplicative constant),

$$(5.74) \quad A_1(b) = \frac{(b^2 + b + 1)(b^2 + 4b + 1)^2}{(1 - b)(b + 1)^5}$$

To leading order in δ , this also solves the Picard-Fuchs equation $\mathcal{D}_2\varpi = 0$. At next-to-leading order, we find that $\mathcal{D}_1\varpi = 0$ implies that A_2 satisfies a non-homogenous differential equation similar to (5.73), with a right-hand side determined by A_1 . The equation $\mathcal{D}_2\varpi = 0$ uniquely determines the homogenous solution, leading to

$$(5.75) \quad A_2(b) = \frac{(b^2 + b + 1)^2 (b^2 + 4b + 1)^3}{2(b - 1)^7 (b + 1)^{11}} p_8(b)$$

$$p_8(b) = b^8 - 18b^7 - 32b^6 - 74b^5 - 114b^4 - 74b^3 - 32b^2 - 18b + 1$$

This procedure may be continued at higher order, for example

$$(5.76) \quad A_3(b) = -\frac{(b^2 + b + 1)^3 (b^2 + 4b + 1)^4}{3(b - 1)^{13} (b + 1)^{17}} p_{16}(b)$$

$$(5.77) \quad p_{16}(b) = b^{16} - 6b^{15} + 508b^{14} + 2930b^{13} + 11572b^{12} + 31310b^{11} + 60840b^{10} \\ + 90470b^9 + 103710b^8 + 90470b^7 + 60840b^6 + 31310b^5 + 11572b^4 \\ + 2930b^3 + 508b^2 - 6b + 1$$

In the second case, plugging in the Ansatz

$$(5.78) \quad \varpi_3^c = \varpi_1^c \log \delta + G_0(b) + G_1(b)\delta + G_2(b)\delta^2 + \dots$$

we find we find that $G_0(b)$ satisfies an inhomogenous second order differential equation,

$$(5.79) \quad b^2 (b^4 + b^3 - b - 1) \partial_b^2 G_0 + b (b^4 - 4b^3 - 8b^2 - 6b - 1) \partial_b G_0 = (b - 1)^4$$

The homogeneous solutions are linear combinations of 1 and $\log(-\frac{b(b^2+b+1)^3}{(b+1)^8})$, which are just the usual constant periods 1 and m . A particular inhomogeneous solution, valid in the half-plane $\Re(b) > 0$, is given by

$$(5.80) \quad G_0(b) = -[\text{Li}_2(b^3) - 4\text{Li}_2(b^2) + 5\text{Li}_2(b)] - \frac{1}{2}(\log b)^2 - 3\log b \log(b^2 + b + 1) + 8\log b \log(b + 1)$$

For $\Re(b) < 0$, this expression is no longer a solution for the standard choice of branch cuts for the logarithm and dilogarithm, nor is it a continuous function of b , but this may be repaired by adding a term proportional to $\log b$ with piecewise constant coefficient, along with a piecewise constant,

$$(5.81) \quad \tilde{G}_0(b) = G_0(b) - 6\pi i \Theta(-b_1 - \frac{1}{2}) \text{sign}(b_2) \left(1 - \Theta(b_2 - \sqrt{3}b_1) \Theta(-b_2 - \sqrt{3}b_1) \Theta(b_2)\right) \log b \\ - 2\pi^2 \Theta(-b_1) \left[2\Theta(|b| - 1) \Theta(b_2 - \sqrt{3}b_1) \Theta(-b_2 - \sqrt{3}b_1) + \text{sign}(1 - |b|)\right] + \frac{\pi^2}{3}$$

where $b = b_1 + ib_2$ and $\Theta(x)$ is the Heaviside function, equal to 1 for $x > 0$ and 0 otherwise. With this definition, $\tilde{G}_0(b)$ is continuous for all $b \in \mathbb{C} \setminus (-\infty, 0) \cup \{\pm e^{i\pi/6}\}$, and satisfies the

differential equation (5.79). In Appendix B, we shall rederive this expression in a completely different way, by computing the regulator period of the family of nodal curves.

At the next order, we find that both Picard-Fuchs equations produce the same inhomogeneous first order ODE for $G_1(b)$, where the homogenous part is the same as for $A_1(b)$ (5.73),

$$(5.82) \quad \begin{aligned} & (b^6 + 5b^5 + 5b^4 - 5b^2 - 5b - 1) G_1'(b) + (5b^4 + 4b^3 + 18b^2 + 4b + 5) G_1(b) = f_1(b) \\ & f_1(b) = \frac{3(b^2 + b + 1)(b^2 + 4b + 1)^2(b^6 + 8b^5 + 15b^4 + 24b^3 + 15b^2 + 8b + 1)}{(b-1)b(b+1)^5} \end{aligned}$$

The inhomogenous solution is given by

$$(5.83) \quad G_1(b) = -\frac{(b^2 + b + 1)(b^2 + 4b + 1)^2}{(b-1)(b+1)^5} \log \left(\frac{b^3(b^2 + b + 1)(b^2 + 4b + 1)^2}{(1-b^2)^6} \right)$$

Let us now express the periods in terms of the modular parameter τ , which lies at a cusp $\tau_c \in \mathbb{R}$ as $u \rightarrow u_c$. For this it is convenient to introduce a new modular parameter τ' , related to τ by an element of $PGL(2, \mathbb{Z})$ such that the limit $\tau = \tau_c$ becomes $\tau' = i\infty$. To relate δ and τ' , it suffices to equate the Laurent expansion of the Klein Hauptmodul $J(u_c + \delta, \ell)$ as $\delta \rightarrow 0$ with the q' -expansion of $J(\tau)$. In this way, we find

$$(5.84) \quad \delta = -\frac{(b^2-1)^6}{b^3(b^2+b+1)(b^2+4b+1)^2} q' + \frac{2(b^2-1)^6(4b^8+37b^7+106b^6+199b^5+244b^4+199b^3+106b^2+37b+4)}{b^6(b^2+b+1)(b^2+4b+1)^3} q'^2 + \dots$$

XXX

To obtain the expansion near a zero u_c of Δ_4 , corresponding to $\tau = \tau_c$, let us introduce a local coordinate τ' , related to τ by an element of $PGL(2, \mathbb{Z})$ which maps $\tau = \tau_c$ to $\tau' = i\infty$. We can then expand the Klein J-invariant near u_c , and equate it with the expansion of $J(\tau')$ near $i\infty$. In practice, we introduce the ratio

$$(5.85) \quad \begin{aligned} P(u) := \frac{\Delta_4(u)}{u - u_c} = & \ell^3 (4u_c^3 \ell (16\ell - 27) - 108u_c^2 \ell - 16u_c \ell + 1) \\ & + 2\ell^4 (u - u_c) (u_c^2 (48\ell - 81) - 54u_c - 4) + 4\ell^4 (u - u_c)^2 (u_c (16\ell - 27) - 9) \\ & + \ell^4 (16\ell - 27) (u - u_c)^3 \end{aligned}$$

which is a cubic polynomial whenever $\Delta_4(u_c) = 0$, and Laurent expand $J(u) = \frac{(Q_4(u))^3}{u^8(u-u_c)P(u)}$ near u_c . In this way we arrive at

$$(5.86) \quad u = u_c + \frac{(16u_c^4 \ell^2 - 24u_c^3 \ell - 8u_c^2 \ell + 1)^3}{u_c^8 \ell^3 (4u_c^3 \ell (16\ell - 27) - 108u_c^2 \ell - 16u_c \ell + 1)} q' + \mathcal{O}(q'^2)$$

where higher order corrections are readily computed. For $\ell = \ell$, the conifold points are given by the roots of a quadratic polynomial, and the expansion can be computed more directly,

$$(5.87) \quad u = \frac{8}{153} \left(5 \pm 8i\sqrt{2} \right) - \frac{2048i(\pm 11369\sqrt{2} - 6976i)}{5688387} q' + \frac{16384(1680185024 \pm 3039785485i\sqrt{2})}{1903396862457} q'^2 + \dots$$

5.4. Limit to local $\mathbb{P}^2$. As $\lambda \rightarrow 0$ keeping u fixed, the Klein Hauptmodul reduces to

$$(5.88) \quad J \rightarrow \frac{(1 - 24u^3)^3}{u^9(1 - 27u^3)} = 1728 + \frac{(1 - 36u^3 + 216u^6)^2}{u^9(1 - 27u^3)}$$

This agrees with [1, A.7] upon identifying $z' = -u^3$ hence $z = u^3 - \frac{1}{27}$. The algebraic Yukawa coupling (5.56) also reduces to the algebraic Yukawa coupling of local $\mathbb{P}^2$,

$$(5.89) \quad C_{uuu}(du)^3 \rightarrow -\frac{(dz)^3}{3z^3(1 + 27z)}$$

The A-periods similarly reduce to those of local $\mathbb{P}^2$, up to an overall factor,

$$(5.90) \quad \begin{aligned} -\omega_1 &\rightarrow 1 - 6z + 90z^2 - 1680z^3 + \dots \\ 3\varpi_2 &\rightarrow \log z - 6z + 45z^2 - 560z^3 + \dots \end{aligned}$$

For the B-periods, we first need to apply a flop transition...

5.5. Monodromies. The spectral flow (2.2) acts on the period vector $\Pi = (1, m, T, T_D)^t$ by

$$(5.91) \quad \Pi \mapsto \begin{pmatrix} 1 & 0 & 0 & 0 \\ m_B - \frac{2m_F}{3} & 1 & 0 & 0 \\ \frac{m_F}{3} & 0 & 1 & 0 \\ m_B m_F - \frac{m_B^2}{2} & m_F - m_B & m_B + 2m_F & 1 \end{pmatrix} \Pi = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ k & 0 & 1 & 0 \\ 4k^2 & k & 8k & 1 \end{pmatrix} \Pi$$

where in the second equality, we specialized to $(m_F, m_B) = k(3, 2)$, which leaves m invariant and sends $T \mapsto T + 8k$. Moreover, the *inverse* of the monodromy matrix associated to a Seidel-Thomas twist with respect to $\mathcal{O}(m_F, m_B)$ acts by

$$(5.92) \quad M_{(m_F, m_B)} := \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2}m_B(m_B - 2m_F) & m_F - m_B & m_B + 2m_F + 1 & -1 \\ \frac{1}{2}m_B(m_B^2 - 4m_F^2) & -m_B^2 - m_B m_F + 2m_F^2 & (m_B + 2m_F)^2 & -m_B - 2m_F + 1 \end{pmatrix}$$

such that $\mathcal{O}(m_F, m_B)$ becomes massless at

$$(5.93) \quad \tau = 2m_F + m_B, \quad T_D = (2m_F + m_B)T + m(m_F - m_B) + \frac{1}{2}m_B^2 - m_B m_F$$

Using the sagemath package `ore_algebra.analytic.monodromy` [35], we find the following monodromies:

- For $\lambda = 1$, around the roots $u_1, \dots, u_4$ of Δ' , ordered as $\{0.2632, -3.0257, -0.2551 + 0.2216i, -0.2551 - 0.2216i\}$

$$(5.94) \quad \begin{aligned} M_1 &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, & M_2 &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & -2 & -4 & 3 \end{pmatrix}, \\ M_3 &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & -2 & 1 \\ \frac{3}{2} & 0 & -9 & 4 \end{pmatrix}, & M_4 &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & 4 & 1 \\ -\frac{3}{2} & 0 & -9 & -2 \end{pmatrix} \end{aligned}$$

Note in particular that

$$(5.95) \quad M_3 \cdot M_2 \cdot M_1 \cdot M_4 = M_{LV} := \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 4 & 1 & 8 & 1 \end{pmatrix}$$

Using (5.92), the corresponding massless objects are $\mathcal{O}(0, 0)$, $\mathcal{O}(1, 0)$, $\mathcal{O}(1, 1)$ and $\mathcal{O}(-1, -1)$, respectively, which become massless at $\tau = 0, -2, 3, -3$. **Actually, it seems that $\mathcal{O}(1, 0)$ should be replaced by $GV(0, 1, \frac{1}{2})$ while $\mathcal{O}(-1, -1)$ should be replaced by its image under monodromy $\mathcal{O}(2, 1)$. The objects above do seem to arise at $\lambda = 2$, however.**

- For $27 + 256\lambda^3 = 0$, specifically $\lambda = \underline{\lambda} := -\frac{3}{4 \cdot 2^{2/3}}$ we find instead 4 monodromy matrices, M_{LV} along with

(5.96)

$$M'_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 & 1 \\ -\frac{1}{2} & 1 & -1 & 1 \end{pmatrix}, \quad M'_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & -2 & -4 & -1 \end{pmatrix}, \quad M'_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & -2 & 1 \\ \frac{3}{2} & 0 & -9 & 4 \end{pmatrix}$$

around the points

(5.97)

$$u'_1 = -\frac{8\lambda}{9} \simeq 0.419974, \quad u'_2 = \frac{8\lambda}{153}(5 - 8i\sqrt{2}) \simeq -0.123522 - 0.279498i, \quad u'_3 = \frac{8\lambda}{153}(5 + 8i\sqrt{2}),$$

such that

$$(5.98) \quad M'_3 M'_1 M'_2 M_{LV} = 1, \quad (M'_1)^6 = 1, \quad (M'_1)^3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

The point u'_1 is an Argyres-Douglas type point, where both $\mathcal{O}(0,0)$ and $\mathcal{O}(0,1)$ are massless, since of $T = m - \frac{1}{2}, T_D = 0$ at that point. In fact, the inverse of M'_1 is the product of two ST twists, $(M'_1)^{-1} = M_{(0,0)} \cdot M_{(0,1)}$. **More to the point, $(M'_1)^{-1} = M_{GV(0,1,-\frac{1}{2})} \cdot M_{(0,0)}$.** On the other hand, u'_2 and u'_3 are conifold points where $\mathcal{O}(-1,0)$ and $\mathcal{O}(1,1)$ are massless, respectively.

- For $\lambda = -1/4$, $m = \frac{\log(1/4)}{2\pi i} + \frac{1}{2}$, we find

$$(5.99) \quad \begin{aligned} M''_1 &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\frac{1}{2} & 1 & -1 & 1 \end{pmatrix}, & M''_2 &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & -2 & -4 & -1 \end{pmatrix}, \\ M''_3 &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & -2 & 1 \\ \frac{3}{2} & 0 & -9 & 4 \end{pmatrix}, & M''_4 &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

correspond to SD twists with respect to $[0,0,1,-\frac{1}{2}]$, $\mathcal{O}(-1,0)$, $\mathcal{O}(1,1)$, $\mathcal{O}(0,0)$, which are massless at $\tau = i\infty, 3, 1, -1$, respectively. Note that $M'_1 = M''_4 M''_1$, $M'_2 = M''_2$, $M'_3 = M''_3$, $M'_1 = M''_4$.

- For $\lambda = -(-1)^{1/3}/4 \simeq \frac{1}{8} + .216i$, the four monodromies are associated to $\mathcal{O}(0,0)$, $\mathcal{O}(1,1)$, $\mathcal{O}(-1,-1)$ and $GV(0,1,-\frac{1}{2})$. The product of the last two gives the monodromy of the AD point.
- For $\lambda = -(-1)^{2/3}/4 \simeq \frac{1}{8} - .216i$, the four monodromies are associated to $\mathcal{O}(0,0)$, $\mathcal{O}(-1,-1)$, $\mathcal{O}(1,1)$ and $GV(0,1,\frac{1}{2})$. The product of the last two gives the monodromy of the AD point.

Finally, at the orbifold point $u = -8\lambda/9$ for λ real, we seem to find

$$(5.100) \quad \Re T_D = -2\Re T, \quad \Im T = -\frac{1}{2}\Im m, \quad \Re \tau = -\frac{1}{4}$$

while $\Im T_D$ and $\Re T$ seem to be random values.

Redoing the analysis in (u, ℓ) coordinates, we find

- For $\ell = 1$, around the roots $u_1, \dots, u_4$ for Δ_4 , ordered as $\{-0.255099+0.221553i, -0.255099-0.221553i, -3.02572, 0.263185\}$

$$(5.101) \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & -2 & 1 \\ \frac{3}{2} & 0 & -9 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & 4 & 1 \\ -\frac{3}{2} & 0 & -9 & -2 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{1}{2} & 1 & -1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

corresponding to spherical twist with respect to $\mathcal{O}(1, 1), \mathcal{O}(-1, -1), GV[0, 1, 1/2], \mathcal{O}(0, 0)$, respectively.

- For $\ell = 2$, around the roots $u_1, \dots, u_4$ for Δ_4 , ordered as $\{0.2, 7.41375, -0.206875 + 0.156987i, -0.206875 - 0.156987i\}$

$$(5.102) \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & -2 & -4 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & -2 & 1 \\ \frac{3}{2} & 0 & -9 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & 4 & 1 \\ -\frac{3}{2} & 0 & -9 & -2 \end{pmatrix}$$

corresponding to spherical twist with respect to $\mathcal{O}(0, 0), \mathcal{O}(1, 0), \mathcal{O}(1, 1), \mathcal{O}(-1, -1)$, respectively.

- For $\ell = -1$, around the roots $u_1, \dots, u_4$ for Δ_4 , ordered as $\{-0.512108+0.1945i, -0.512108-0.1945i, 0.0935038 + 0.262211i, 0.0935038 - 0.262211i\}$,

$$(5.103) \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{3}{2} & -1 & -4 & 1 \\ \frac{15}{2} & -5 & -25 & 6 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & -2 & -4 & -1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & -2 & 1 \\ \frac{3}{2} & 0 & -9 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

corresponding to spherical twist with respect to $\mathcal{O}(2, 1), \mathcal{O}(-1, 0), \mathcal{O}(1, 1), \mathcal{O}(0, 0)$, respectively.

- For $\ell = -27/256$, more precisely $m = -1/6 + 0.119331i$ around the 3 roots $u_1, \dots, u_3$ of Δ_4 ordered as

$$(5.104) \quad u_1 = -\frac{8}{9}, \quad u_2 = \frac{8}{153}(5 + 8i\sqrt{2}) \simeq 0.261 + 0.592i, \quad u_3 = \frac{8}{153}(5 - 8i\sqrt{2}) \simeq 0.261 - 0.592i,$$

$$(5.105) \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & -2 & 1 \\ \frac{3}{2} & 1 & -7 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & 4 & 1 \\ -\frac{3}{2} & 0 & -9 & -2 \end{pmatrix}$$

The last two correspond to spherical twists with respect to $\mathcal{O}(0, 0)$ and $\mathcal{O}(-1, -1)$, while the first one, associated to the Argyres-Douglas point, is a monodromy of order 6, equal to the inverse of $M_{\mathcal{O}(1,1)} \cdot M_{GV(0,1,1/2)}$. The values of (τ, T, T_D) at this point are $(\frac{5}{2} + i\frac{\sqrt{3}}{2}, m + \frac{1}{2}, 3m + 1)$, invariant under monodromy as they should.

APPENDIX A. THE NON-CONGRUENCE SUBGROUP Γ

The monodromy group for $\lambda = \frac{\lambda}{\lambda} := -\frac{3}{4^{2/3}}$ is an extension of a subgroup of $PSL(2, \mathbb{Z})$ generated by the lower-right blocks of the monodromy matrices (5.96),

$$(A.1) \quad g_1 = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \quad g_2 = \begin{pmatrix} -1 & -4 \\ 1 & 3 \end{pmatrix}, \quad g_3 = \begin{pmatrix} -1 & -1 \\ 4 & 3 \end{pmatrix},$$

such that

$$(A.2) \quad g_1^3 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad g_1 \cdot g_3 \cdot g_2 = \begin{pmatrix} 1 & 8 \\ 0 & 1 \end{pmatrix}, \quad g_1 \cdot g_3 \cdot g_1^{-1} = \begin{pmatrix} 4 & -9 \\ 1 & -2 \end{pmatrix}$$

As anticipated in [16] on the basis of the classification of rational elliptic surfaces, this is a non-congruence subgroup of $PSL(2, \mathbb{Z})$ of index 10, genus zero, with 3 cusps of widths $(8, 1, 1)$, no elliptic point of order 2, and one elliptic point of order 3 (corresponding to the Argyres-Douglas point). Non-congruence subgroups of index ≤ 11 were classified in [34], and a conjugate of Γ , denoted by $\Gamma' = \Gamma(10; 0, 3, 0, 1; 8)$, appears as the last in the index 10 subgroups in Table 2. As noted in [34, Example 34], Γ' has width one cusps at $\tau = 2$ and 5 modulo 8, whereas Γ has width one cusps at $\tau = 3$ and $\tau = 6$ modulo 8, so $\Gamma = T\Gamma'T^{-1}$.

In general, subgroups of $PSL(2, \mathbb{Z})$ are classified by the permutations σ_S, σ_R through which the order two and three generators $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, R = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$ act on the cosets $PSL(2, \mathbb{Z})/\Gamma$. Relabelling those cosets such that the generator $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ acts by cyclic permutations of the first 8 of them, $\sigma_T = (1, 2, 3, 4, 5, 6, 7, 8)(9)(10)$, we have

$$(A.3) \quad \sigma_S = (1, 8)(2, 4)(5, 7)(3, 9)(6, 10), \quad \sigma_R = (8, 2, 5)(4, 3, 9)(7, 6, 10)$$

Using Sagemath package `sage.modular.arithgroup.arithgroup_perm`⁹, we readily check that Γ is generated by the matrices in (A.1), and obtain an explicit list of cosets:

$$(A.4) \quad \{c_i\}_{i=1\dots 10} = \{1, T^7, T^6, T^5, T^4, T^3, T^2, T, \begin{pmatrix} 6 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}\}$$

A fundamental domain of Γ can thus be obtained as the union of the images of the standard ‘keyhole’ fundamental domain of $PSL(2, \mathbb{Z})$ under the cosets c_i (see Figure 9).

Note that the generators g_i in (A.1) leave the points $\tau = -2, -\frac{1}{2}, e^{i\pi/3}$ invariant, respectively. Under these generators, we find the orbit of the three cusps and the elliptic point in the range $0 \leq \tau_1 < 8$,

$$(A.5) \quad \{i\infty, 0, 1, 2, 4, 5, 7, \frac{1,7,9,13}{2}, \frac{4,5,7,8,10,11,13,14,16,17,19,20}{3}, \dots\},$$

$$(A.6) \quad \{3, \frac{15}{2}, \frac{2}{3}, \dots\}, \quad \{6, \frac{3}{2}, \frac{1}{3}, \dots\}, \quad \{e^{i\pi/3}, \frac{9}{2} + \frac{i}{2\sqrt{3}}, \frac{175+i\sqrt{3}}{26}, \dots\}$$

The dots indicate images with larger denominators, or, in the case of the elliptic point, smaller imaginary parts. In particular, the fixed points $\tau = -2, \tau = -1/2$ under g_1, g_2 are in the orbit of $\tau = 6$ and $\tau = 3$, respectively, and these two orbits are exchanged under $\tau \mapsto 1 - \bar{\tau}$.

Modular forms for the non-congruence subgroup Γ' were computed in [36, 37]¹⁰. The generators provided by Sage,

$$(A.7) \quad g'_1 = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}, \quad g'_2 = \begin{pmatrix} 3 & -1 \\ 4 & -1 \end{pmatrix}, \quad g'_3 = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$$

satisfy

$$(A.8) \quad (g'_1)^3 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad g'_3 g'_2 g'_1{}^{-1} = \begin{pmatrix} 1 & 8 \\ 0 & 1 \end{pmatrix},$$

and are related to those in (A.1) by

$$(A.9) \quad g'_1 = T^{-1} g_1^{-1} T, \quad g'_2 = T^{-1} g_1 g_2 g_1^{-1} T, \quad g'_3 = T^{-1} g_1 g_3 g_1^{-1} T.$$

confirming that Γ' is therefore conjugate to Γ under T . In particular, Γ' has cusps of width 1 at $\tau = 2, 5$, and width 8 at $\tau = i\infty$. The Hauptmodul $J_{\Gamma'}$ was determined in [37] via the Belyi

⁹Define `G = ArithmeticSubgroup_Permutation(S2="(1,8)(2,4)(5,7)(3,9)(6,10)", S3="(8,2,5)(4,3,9)(7,6,10)")` and type `G.generators()`, `G.cosets()`

¹⁰The unique LMFDB identifier is 10T30-8.1.1-3.3.3.1-2.2.2.2.2-a, where 10T30 stands for the finite group $PGL(2, \mathbb{F}_9) = A_6 \rtimes C_2$ spanned by σ_S and σ_R of order 720 inside the permutation group S_{10} .

map relating it to the Klein J-invariant,

$$(A.10) \quad J(\tau) = \frac{(10w + J_{\Gamma'}) (208w^3 + 120w^2 J_{\Gamma'} - 6w J_{\Gamma'}^2 + J_{\Gamma'}^3)^3}{J_{\Gamma'}^2 - 8w J_{\Gamma'} + 144w^2} \\ = 1728 + \frac{(70624w^5 - 760w^4 J_{\Gamma'} + 928w^3 J_{\Gamma'}^2 + 136w^2 J_{\Gamma'}^3 - 4w J_{\Gamma'}^4 + J_{\Gamma'}^5)^2}{J_{\Gamma'}^2 - 8w J_{\Gamma'} + 144w^2}$$

where $w := -(-1/3^9)^{1/8} \simeq -0.268444 - 0.111193i$. Inserting the q -expansion $J(\tau) = 1/q_8^8 + 744 + \dots$, where $q_8 := e^{2\pi i \tau/8}$, one obtains the q_8 -series expansion

$$(A.11) \quad J_{\Gamma'} = \frac{1}{q_8} - 18q_8 w^2 - 240q_8^2 w^3 + 288q_8^3 w^4 - 2304q_8^4 w^5 - 10176q_8^5 w^6 + 31680q_8^6 w^7 \\ - 111987q_8^7 w^8 + 243712q_8^8 w^9 + 712170q_8^9 w^{10} + 19958112q_8^{10} w^{11} - 28207584q_8^{11} w^{12} + \dots$$

Note that after rescaling $q_8 \rightarrow q_8/w$, the Fourier coefficients are integer up to an overall factor of w .

As explained in [36, §5.3], a basis of holomorphic modular forms of (even) weight k for a subgroup Γ with Hauptmodul J can be constructed from powers of the τ -derivative $q\partial_q J$, multiplied by a suitable rational function in J . Indeed, $q\partial_q J$ transforms as a modular form of weight 2 and vanishes at finite cusps and at elliptic points. Denoting by $\mathcal{C}$, $\mathcal{E}_2$ and $\mathcal{E}_3$ the set of finite cusps, elliptic points of order 2 and elliptic points of order 3, the following ratio is then a holomorphic modular form:

$$(A.12) \quad \phi_{2k,i} = \frac{(q\partial_q J)^{k/2} J^{d-k/2-i}}{\prod_{e \in \mathcal{E}_2} (J - j_e)^{\lfloor k/4 \rfloor} \prod_{e \in \mathcal{E}_3} (J - j_e)^{\lfloor k/3 \rfloor} \prod_{c \in \mathcal{C}} (J - j_c)^{k/2}}$$

with $d = |\mathcal{E}_2| \lfloor k/4 \rfloor + |\mathcal{E}_3| \lfloor k/3 \rfloor + (|\mathcal{C}| - 1) \frac{k}{2}$ is the degree of the denominator and i runs over the range $0 \leq i \leq d - k/2$. For the group Γ' with one order 3 elliptic point and two cusps, this reduces to

$$(A.13) \quad \phi_{2k,i} = \frac{(q_8 \partial_{q_8} J_{\Gamma'})^{k/2} J_{\Gamma'}^{\lfloor k/3 \rfloor + k/2 - i}}{(J_{\Gamma'} + 10w)^{\lfloor k/3 \rfloor} (J_{\Gamma'}^2 - 8w J_{\Gamma'} + 144w^2)^{k/2}}$$

for $0 \leq i \leq \lfloor k/3 \rfloor + k/2$, leading to a two-dimensional space of modular forms of weight 2, respectively 4-dimensional space of weight 4, 6-dimensional space of weight 6, etc. A basis of cusp forms can similarly be obtained by requiring a factor $(\prod_{c \in \mathcal{C}} (J - j_c))/J$ in the numerator,

$$(A.14) \quad \psi_{2k,i} = \frac{(q_8 \partial_{q_8} J_{\Gamma'})^{k/2} J_{\Gamma'}^{\lfloor k/3 \rfloor + k/2 - i - 2}}{(J_{\Gamma'} + 10w)^{\lfloor k/3 \rfloor} (J_{\Gamma'}^2 - 8w J_{\Gamma'} + 144w^2)^{\frac{k}{2} - 1}}$$

for $0 \leq i \leq \lfloor k/3 \rfloor + k/2 - 3$. The first cusp form $\psi_{4,0}$ thus occurs in weight 4.

For $\lambda = \underline{\lambda}$, we find that the q_8 -expansion of $1/u$ in (5.53) matches the q'_8 expansion (A.11) of the Hauptmodul of Γ' , upon replacing $q_8 \mapsto e^{-i\pi/4} q'_8$ (taking into account that Γ and Γ' are conjugate under T), up to an additive constant and multiplicative factor,

$$(A.15) \quad 1/u = \frac{J_{\Gamma'} + w}{8\underline{\lambda}w}$$

Indeed, under this identification, (A.10) reduces to

$$(A.16) \quad J(\tau) = - \frac{(8\underline{\lambda} + 9u) (81u^3 + 1080\underline{\lambda}u^2 - 576\underline{\lambda}^2u - 54)^3}{3^9 u^8 (153u^2 - 80\underline{\lambda}u + 64\underline{\lambda}^2)} \\ = 1728 - \frac{(72171u^5 - 17496\underline{\lambda}u^4 + 31104\underline{\lambda}^2u^3 - 8748u^2 - 36864\underline{\lambda}^4u + 32768\underline{\lambda}^5)^2}{3^9 u^8 (153u^2 - 80\underline{\lambda}u + 64\underline{\lambda}^2)}$$

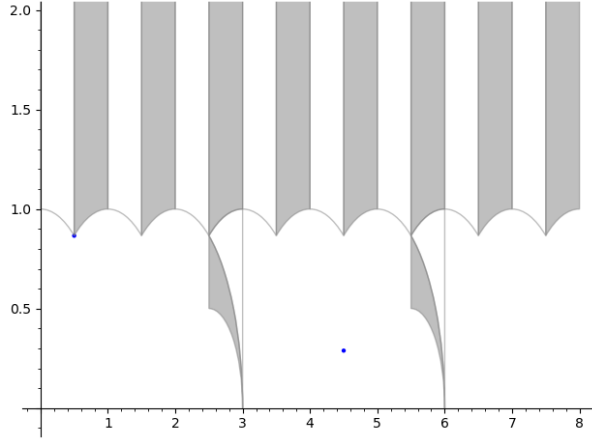


FIGURE 9. A fundamental domain for the non-congruence subgroup Γ . The blue dots correspond to elliptic points of order 3.

which matches the Klein Hauptmodul (5.10) at $\lambda = \underline{\lambda}$, using the fact that

(A.17)

$$1 - 8\underline{\lambda}u^2 - 24u^3 + 16\underline{\lambda}^2u^4 = -\frac{(9u + 8\underline{\lambda})}{432\underline{\lambda}} (81u^3 + 1080u^2\underline{\lambda} - 576u\underline{\lambda}^2 - 54)$$

$$\underline{\lambda} + u - 8\underline{\lambda}^2u^2 - 36\underline{\lambda}u^3 + (16\underline{\lambda}^3 - 27)u^4 = -\frac{(9u + 8\underline{\lambda})^2}{432} (153u^2 - 80u\underline{\lambda} + 64\underline{\lambda}^2)$$

Moreover, the square of the inverse Yukawa $C^2 = (1/C_{TTT})^2$ can be identified as a weight 6 holomorphic modular form, see (5.64), while the square of the A-period $(\omega_1)^2$ is identified as a weight 2 meromorphic modular form with a simple pole at the orbifold point, see (5.65).

Away from the locus $\lambda = \underline{\lambda}$, it seems that one should express u in terms of the Hauptmodul $\underline{u} = \frac{8\lambda w}{J_{\Gamma'} + w}$ by equating the numerators and denominators, up to a scale factor f ,

(A.18)

$$1 - 8\lambda u^2 - 24u^3 + 16\lambda^2u^4 = -f \frac{(9\underline{u} + 8\underline{\lambda})}{432\underline{\lambda}} (81\underline{u}^3 + 1080\underline{u}^2\underline{\lambda} - 576\underline{u}\underline{\lambda}^2 - 54)$$

$$u^8 (\lambda + u - 8\lambda^2u^2 - 36\lambda u^3 + (16\lambda^3 - 27)u^4) = -f^3 \underline{u}^8 \frac{(9\underline{u} + 8\underline{\lambda})^2}{432} (153\underline{u}^2 - 80\underline{u}\underline{\lambda} + 64\underline{\lambda}^2)$$

Solving order by order in powers of $\lambda - \underline{\lambda}$, we find

(A.19)

$$u = \underline{u} - \frac{(\underline{u} (32\underline{\lambda}^2 \underline{u} (891\underline{u}^6 - 126\underline{u}^3 + 4) + 72\underline{\lambda} \underline{u}^2 (15\underline{u}^3 - 2) + 3(243\underline{u}^6 - 90\underline{u}^3 + 4)))}{6(192\underline{\lambda}^2(12\underline{u}^3 - 1)\underline{u}^2 + 4\underline{\lambda}(891\underline{u}^6 - 144\underline{u}^3 + 4) - 81\underline{u}^4)} (\lambda - \underline{\lambda})$$

$$+ O((\lambda - \underline{\lambda})^2)$$

$$f = 1 - \frac{12(16\underline{\lambda}^2(891\underline{u}^6 - 144\underline{u}^3 + 4)\underline{u}^2 - 324\underline{\lambda}\underline{u}^6 - 972\underline{u}^7 + 81\underline{u}^4)}{(8\underline{\lambda} + 9\underline{u})(192\underline{\lambda}^2(12\underline{u}^3 - 1)\underline{u}^2 + 4\underline{\lambda}(891\underline{u}^6 - 144\underline{u}^3 + 4) - 81\underline{u}^4)} (\lambda - \underline{\lambda})$$

$$+ O((\lambda - \underline{\lambda})^2)$$

$$= 1 - \frac{12(16\underline{\lambda}^2(891\underline{u}^6 - 144\underline{u}^3 + 4)\underline{u}^2 - 324\underline{\lambda}\underline{u}^6 - 972\underline{u}^7 + 81\underline{u}^4)}{(8\underline{\lambda} + 9\underline{u})(192\underline{\lambda}^2(12\underline{u}^3 - 1)\underline{u}^2 + 4\underline{\lambda}(891\underline{u}^6 - 144\underline{u}^3 + 4) - 81\underline{u}^4)} (\lambda - \underline{\lambda}) + O((\lambda - \underline{\lambda})^2)$$

The value of J_{Γ} where $j = j_B$ is given by

$$(A.20) \quad J_{\Gamma}/w = -10 - \frac{2^{52}}{3^{17}}(\ell - \underline{\ell}) - \frac{2^{57} \cdot 103 \cdot 199 \cdot 401 \cdot 857 \cdot 871621}{3^{38}}(\ell - \underline{\ell})^2 + \dots$$

APPENDIX B. CONIFOLD PERIODS FROM DILOGARITHMS

In [15], a formalism was developed for computing the regulator period of singular elliptic curves in toric ambient spaces. In the simplest case of a type I_1 nodal degeneration, the key step is to obtain a rational parametrization of the nodal curve of the form

$$(B.1) \quad X = A \prod_j (1 - \alpha_j/z)^{d_j}, \quad Y = B \prod_k (1 - z/\beta_k)^{e_k}$$

such that the node connects the points $z = 0$ and $z = \infty$. The regulator period is then given by [15, (4.12)]

$$(B.2) \quad \begin{aligned} \mathcal{Q} = & \frac{1}{2\pi} \sum_{j,k} d_j e_k \left[\text{Li}_2\left(\frac{\alpha_j}{\beta_k}\right) + (\log \alpha_j - \log \beta_k) \log\left(1 - \frac{\alpha_j}{\beta_k}\right) \right] \\ & - \frac{1}{2\pi} \log Y(0) (\log X(0) - \log X(\infty)) + \log Y(0) \sum_j d_j \log \alpha_j \\ & - \log X(\infty) \sum_k e_k \log \beta_k. \end{aligned}$$

Note that $\sum_j d_j = \sum_k e_k = 0$ in order for X, Y to take finite values at $z = 0, \infty$.

Here we apply this formalism to compute the values of T and T_D at conifold points, where the mirror curve admits a nodal degeneration.¹¹ We start with the mirror curve in toric coordinates

$$(B.3) \quad x + y + \frac{\ell}{xy} + \frac{\ell}{x} = \frac{1}{u}$$

In order to put the curve in Weierstrass form (5.4), we perform the birational change of coordinates

$$(B.4) \quad x = \frac{\frac{Y}{12u} + 18\ell u^2}{X + 12\ell u^2 - 3}, \quad y = -\frac{36\ell u^2}{X + 12\ell u^2 - 3}.$$

Now, when $\Delta_4(u) = 0$, the Weierstrass form (5.4) degenerates into

$$(B.5) \quad Y^2 = 4(X - X_0)^2(X - X_1), \quad X_0 = -\frac{3g_3}{2g_2}, \quad X_1 = -2X_0$$

The nodal curve then admits the rational parametrization

$$(B.6) \quad X = t^2 - 2X_0, \quad Y = t(2t^2 - 3X_0).$$

where $t \in \mathbb{C}$, such that the original toric coordinates are given by

$$(B.7) \quad \begin{aligned} x &= \frac{3(t^2 - 2X_0) + t(t^2 - 3X_0) - 9 + 36\ell u^2(1 + 3u)}{6u(t^2 - 2X_0 - 3 + 12\ell u^2)}, \\ y &= \frac{36\ell u^2}{2X_0 + 3 - 12\ell u^2 - t^2}. \end{aligned}$$

Introducing the parameter $t_1 = \sqrt{2X_0 + 3 - 12\ell u^2}$, such that the denominator of y becomes $t^2 - t_1^2$, one verifies that on the discriminant locus, the numerator of x factorizes as

$$(B.8) \quad 3(t^2 - 2X_0) + t(t^2 - 3X_0) - 9 + 36\ell u^2(1 + 3u) = (t - t_1)^2(t + 2t_1 + 3),$$

Moreover, in terms of t_1 we have

$$(B.9) \quad \ell = \frac{27(t_1 + 1)^3(t_1 + 3)}{16t_1^4}, \quad u = \frac{2t_1^2}{9(t_1 + 1)}, \quad X_0 = t_1(t_1 + 2)$$

Going back to the original toric coordinates, we obtain the rational parametrization

$$(B.10) \quad x = \frac{(t - t_1)(t + 2t_1 + 3)}{6u(t + t_1)}, \quad y = -\frac{36u^2}{t^2 - t_1^2}.$$

¹¹We are grateful to Marcos Marino for sharing a note outlining the key steps in this computation.

such that the nodes lie at $t = \pm\sqrt{3X_0}$. In order to map them to 0 and ∞ , it suffices to perform the $SL(2, \mathbb{C})$ transformation

$$(B.11) \quad z = \frac{t - \sqrt{3X_0}}{t + \sqrt{3X_0}}.$$

In terms of z , the parametrization (B.10) takes the form

$$(B.12) \quad x = A \frac{(z - 1/b^2)(z - b)}{(z - 1/b)(z - 1)}, \quad y = B \frac{(z - 1)^2}{(z - b)(z - 1/b)},$$

where

$$(B.13) \quad b = \frac{t_1 - \sqrt{3X_0}}{t_1 + \sqrt{3X_0}}, \quad A = -\frac{b(b^2 + b + 1)}{(b + 1)^4}, \quad B = -\frac{b^2 + b + 1}{(b + 1)^2}$$

We note that on the discriminant locus, u , ℓ and t_1 can all be expressed as rational functions of b ,

$$(B.14) \quad \ell = -\frac{b(b^2 + b + 1)^3}{(b + 1)^8}, \quad u = -\frac{(b + 1)^4}{(b^2 + b + 1)(b^2 + 4b + 1)}, \quad t_1 = -\frac{3(b + 1)^2}{b^2 + 4b + 1}$$

For fixed ℓ , there are generically 8 values of b , related in pairs under the involution $b \rightarrow 1/b$. At the special value $\ell = -27/256$, the root $b = 1$ occurs with multiplicity 4, corresponding to $u = -8/9$, along with two additional pairs of roots corresponding to $u = \frac{8}{153}(5 \pm 8i\sqrt{2})$. At $\ell = 0$, the roots $e^{2\pi i/3}, e^{4\pi i/3}$ each occur with multiplicity 3, while $b = 0$ occurs with multiplicity 1. At $\ell = \infty$, the root $b = -1$ occurs with multiplicity 8. This ramification structure is of course consistent with Riemann-Hurwitz formula (see footnote 5).

Now, applying the formula (B.2) with $\alpha_j = (1, 1, -1, -1), \beta_k = (2, -1, -1)$, and using functional identities satisfied by the dilogarithm function, we arrive at

$$(B.15) \quad \begin{aligned} \mathcal{Q}(b) = & \frac{1}{2\pi} [\text{Li}_2(b^3) - 4\text{Li}_2(b^2) + 5\text{Li}_2(b)] + \frac{1}{4\pi} (\log b)^2 \\ & + \frac{3}{2\pi} \log b \log(b^2 + b + 1) - \frac{4}{\pi} \log b \log(b + 1) - \frac{\pi}{6} \\ & + 3i \Theta(-b_1 - \tfrac{1}{2}) \text{sign}(b_2) \left(1 - \Theta(b_2 - \sqrt{3}b_1) \Theta(-b_2 - \sqrt{3}b_1) \Theta(b_2)\right) \log b \\ & - \pi \Theta(-b_1) \left[2\Theta(|b| - 1) \Theta(b_2 - \sqrt{3}b_1) \Theta(-b_2 - \sqrt{3}b_1) + \text{sign}(1 - |b|)\right] \end{aligned}$$

where $b = b_1 + ib_2$ and $\Theta(x)$ is the Heaviside function. Note that the last two lines vanish when $\Re(b) > 0$, and $\mathcal{Q}(b) = -\tilde{G}_0(b)/(2\pi)$ where $\tilde{G}_0$ was defined in (5.81). One may check that this expression is a continuous function of b , except at the singular points $b = \pm e^{i\pi/6}$ and along the negative semi-axis $(-\infty, 0)$. Moreover, we observe that $\mathcal{Q}$ is an odd function under $b \mapsto 1/b$,

$$(B.16) \quad \mathcal{Q}(b) + \mathcal{Q}(1/b) = 0$$

and that the imaginary part is given by a much simpler formula¹²

$$(B.17) \quad \Im \mathcal{Q}(b) = \frac{1}{2\pi} [\mathcal{D}_2(b^3) - 4\mathcal{D}_2(b^2) + 5\mathcal{D}_2(b)] + \frac{1}{2\pi} \arg(b) \times \log \left| \frac{b(b^2 + b + 1)^3}{(b + 1)^8} \right|$$

where $\mathcal{D}_2(z) = \Im(\text{Li}_2(z)) + \log |z| \arg(1 - z)$ is the Bloch-Wigner dilogarithm. Using the identities

$$(B.18) \quad \mathcal{D}_2(z^2) = 2(\mathcal{D}_2(z) + \mathcal{D}_2(-z)), \quad \mathcal{D}_2(z^3) = 3(\mathcal{D}_2(z) + \mathcal{D}_2(\omega z) + \mathcal{D}_2(\omega^2 z))$$

where $\omega = e^{2\pi i/3}$, we can also write

$$(B.19) \quad \Im \mathcal{Q}(b) = \frac{1}{2\pi} [3\mathcal{D}_2(\omega b) + 3\mathcal{D}_2(\omega^2 b) - 8\mathcal{D}_2(-b)] + \frac{1}{2\pi} \arg(b) \times \log \left| \frac{b(b^2 + b + 1)^3}{(b + 1)^8} \right|$$

¹²The last term proportional to $\log |\ell|$ was predicted by M. Kerr, in private communication with M. Marino.

ℓ	b	u	$\mathcal{Q}(b)/(2\pi)$	T	T_D	massless
1	$-0.7251497610 - 0.6885911879i$ -4.402515245 $-1.016583417 - 1.122079973i$ $-1.016583417 + 1.122079973i$	0.263 -3.026 $-0.255 - 0.222i$ $-0.255 + 0.222i$	$-0.1841935331i$ -0.2309698649 $0.3753225891 - 0.1550612280i$ $0.3753225891 + 0.1550612280i$	$\mathcal{Q}$ $\frac{1}{2}$ $\mathcal{Q}$ $-\mathcal{Q}$	0 $\frac{3}{2} - \mathcal{Q}$ $-3\mathcal{Q} - \frac{1}{2}$ $-3\mathcal{Q} - \frac{1}{2}$	$\mathcal{O}(0, 0)$ $GV(0, 1, \frac{1}{2})$ $\mathcal{O}(-1, -1)$ $\mathcal{O}(1, 1)$
2	-0.2810370499 $-0.7500000000 + 0.6614378278i$ $-1.107886516 - 1.057104151i$ $-1.107886516 + 1.057104151i$	7.413 1/5 $-0.207 - 0.157i$ $-0.207 + 0.157i$	$0.2068883162 + 0.0551589000i$ $0.2263378581i$ $0.3827232997 - 0.1962048569i$ $0.3827232997 + 0.1962048569i$	$-\mathcal{Q} - 2m$ $\mathcal{Q} + m$ $\mathcal{Q} + m$ $\mathcal{Q} - m$	$-2\mathcal{Q} - 3m$ 0 $3\mathcal{Q} + 3m - \frac{1}{2}$ $3\mathcal{Q} - 3m - \frac{1}{2}$	$\mathcal{O}(1, 0)$ $\mathcal{O}(0, 0)$ $\mathcal{O}(1, 1)$ $\mathcal{O}(-1, -1)$
-1	$-0.2575549702 - 0.2778334856i$ $-0.2575549702 + 0.2778334856i$ $-0.6006581337 - 0.5900126542i$ $-0.6006581337 + 0.5900126542i$	$-0.512 - 0.195i$ $-0.512 + 0.195i$ 0.094 + 0.262i 0.094 - 0.262i	$-0.5570091734 - 0.1033985032i$ $-0.5570091734 + 0.1033985032i$ $-0.1891167048 - 0.1775197647i$ $-0.1891167048 + 0.1775197647i$	$\mathcal{Q} + m - 1$ $-\mathcal{Q} + m + 1$ $\mathcal{Q} + m$ $-\mathcal{Q} + m$	$-2\mathcal{Q} - 3m + 2$ $-5\mathcal{Q} + 3m + 4$ 3 $\mathcal{Q}$ 0	$\mathcal{O}(-1, 0)$ $\mathcal{O}(2, 1)$ $\mathcal{O}(1, 1)$ $\mathcal{O}(0, 0)$
$-\frac{27}{256}$	$-0.5459669299 - 0.6836214670i$ $-0.5459669299 + 0.6836214670i$	0.261 + 0.592i 0.261 - 0.592i	$-0.1802984704 - 0.0463640951i$ $-0.1802984704 + 0.0463640951i$	$\mathcal{Q} + m$ $-\mathcal{Q} + m$	0 $3\mathcal{Q} - 3m - \frac{1}{2}$	$\mathcal{O}(0, 0)$ $\mathcal{O}(-1, -1)$

TABLE 3. Periods at conifold points.

We have checked numerically that these formulae correctly reproduce the values of the periods at the conifold point, up to an additive constant and multiple of m (see Table 3).

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